GENERAL SCIENCE GRADE 10: RATES OF REACTIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.5 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.5: Rates of Reactions
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Rates of reactions — meaning and definition of rate of a chemical reaction – Measuring rates of reactions — methods for following reactions (gas volume collected, mass lost, colour change, precipitate formation) – Factors affecting the rate of reactions: temperature – Factors affecting the rate of reactions: concentration – Factors affecting the rate of reactions: surface area – Factors affecting the rate of reactions: catalysts – Factors affecting the rate of reactions: light – Rates of reactions in biological processes (enzyme-controlled reactions, respiration, photosynthesis) – Rates of reactions in chemical and physical processes (industrial processes, food preservation, cooking) – Importance of optimum conditions in biological, chemical and physical processes
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe rate of chemical reactions; b) perform experiments involving the rate of chemical reactions; c) explain factors that influence the rate of reactions in biological, chemical and physical processes; d) appreciate the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life.
Core Competencies	– Communication and Collaboration: as the learner discusses with peers the factors that affect the rate of chemical reactions and presents findings in plenary, including in the context of Kenyan industrial and agricultural processes. – Digital Literacy: as the learner uses digital devices and print media to research on the importance of optimum conditions in biological, chemical and physical processes (e.g., tea processing in Kericho, fish preservation at Lake Victoria). – Critical Thinking and Problem Solving: as the learner designs and interprets experiments to investigate factors affecting reaction rates, identifies variables, and draws evidence-based conclusions. – Learning to Learn: as the learner reflects on experimental outcomes, adjusts procedures, and connects findings to real-life Kenyan contexts such as cooking ugali, fermentation of uji, or rusting of iron sheets.
Core Values	– Responsibility: as the learner carefully uses digital devices to search for information on the importance of optimum conditions in industrial processes, and handles laboratory apparatus safely during reaction-rate experiments.

	– Respect: as the learner respects others' opinions while discussing with peers in a group the importance of efficient industrial processes and the role of reaction rates in daily Kenyan life. – Integrity: as the learner records experimental observations honestly and reports results accurately without falsification. – Unity: as the learner collaborates equitably with group members during practical investigations, sharing roles and responsibilities regardless of gender or background.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner frames investigable questions about why maize porridge (uji) thickens faster at higher temperatures or why food goes bad more quickly in Mombasa's heat than in Nairobi's cooler climate. – Planning and Carrying Out Investigations: as the learner designs controlled experiments to test the effect of temperature, concentration, surface area, and catalysts on reaction rates, identifying independent, dependent, and controlled variables. – Analysing and Interpreting Data: as the learner collects quantitative data (volume of gas produced, time for precipitate to form) and constructs tables and graphs to identify patterns and trends. – Constructing Explanations: as the learner uses particle/collision theory to explain why increasing temperature or concentration speeds up reactions, connecting molecular-level reasoning to observable phenomena. – Engaging in Argument from Evidence: as the learner evaluates experimental results, compares findings with peers, and defends or revises claims about which factors most significantly affect reaction rates. – Obtaining, Evaluating, and Communicating Information: as the learner researches optimum conditions in Kenyan industries (tea processing, cement manufacture, oil refining at Mombasa) and communicates findings using digital and print media.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security: as the learner observes safety measures when carrying out activities to investigate rates of reactions (use of protective equipment, safe handling of dilute acids and hydrogen peroxide, proper disposal of chemical waste). – Financial Literacy: as the learner explores how controlling reaction rates in Kenyan industries (e.g., Bamburi Cement, EABL, tea factories in Kericho) reduces costs, saves energy, and improves product quality, linking chemistry to economic decision-making. – Environmental Issues: as the learner investigates how uncontrolled reaction rates contribute to environmental problems such as acid rain, corrosion of iron-sheet roofing in coastal Mombasa, and food spoilage, and explores mitigation strategies. – Socio-Economic and Environmental Education: as the learner discusses how optimising reaction rates in food processing, medicine production, and agriculture (e.g., use of fertilisers) supports Kenya's economic development and food security goals.
Career Connections	– Industrial Chemist: designing and optimising chemical processes (fertiliser production at Kenya's Athi River, cement manufacture) by controlling reaction rates for maximum efficiency and safety. – Food Scientist / Food Technologist: applying knowledge of reaction rates to food preservation, fermentation (yoghurt, uji, busaa), and value addition of Kenyan agricultural produce such as maize, milk, and cassava. – Pharmacist / Pharmaceutical Scientist: understanding reaction rates in drug formulation, shelf-life determination, and the enzymatic processes underlying medicine efficacy. – Agricultural Extension Officer: advising farmers in the Rift Valley and Central Kenya on how temperature, moisture, and soil chemistry affect nutrient-release rates from fertilisers and organic matter decomposition. – Environmental Scientist: monitoring and managing corrosion, pollution, and degradation processes influenced by reaction kinetics in Kenyan ecosystems including Lake Victoria and coastal marine environments. – Biochemist / Medical Laboratory Scientist: applying enzyme kinetics to clinical diagnostics, metabolic studies, and biotechnology research in Kenyan hospitals and research institutions such as KEMRI.
Focus for Lessons	This unit investigates the rate of chemical reactions and the factors that control how fast reactions proceed in biological, chemical, and physical systems. Learners progress from qualitative observation of everyday Kenyan phenomena — such as why ugali cooks faster on a high flame, why sukuma wiki wilts and decomposes, and why iron-sheet roofs rust faster at the Mombasa coast than in Nairobi — to quantitative experimental investigations of temperature, concentration, surface area, catalysts, and light. The unit

	employs a phenomenon-driven, inquiry-based storyline in which learners build and revise a conceptual model of particle collisions to explain and predict reaction-rate changes, culminating in an evaluation of how Kenyan industries and households can optimise conditions for safety, efficiency, and sustainability.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive? KICD KEY INQUIRY QUESTION: 1. How do various factors affect the rate of reactions?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Disappearing Thiosulfate Cross: A teacher places two identical sheets of white paper, each marked with a bold black cross (×), on a desk at the front of class. Two beakers are set up: both contain dilute hydrochloric acid (HCl), but Beaker A sits in a tray of cold water (approximately 10 °C, simulating a cool Nairobi highland morning) while Beaker B sits in a tray of warm water (approximately 50 °C, simulating afternoon heat in Mombasa or Kisumu). Sodium thiosulfate solution of identical concentration is poured simultaneously into both beakers, which are then placed over the paper crosses. Students observe both beakers from above. In Beaker B (warm), a milky-yellow precipitate of sulfur forms rapidly and the cross disappears from view within about 30–40 seconds. In Beaker A (cold), the cross remains visible for several minutes before it too eventually disappears. Students are immediately puzzled: both beakers contain exactly the same chemicals in the same amounts — so WHY does Beaker B's reaction finish so much faster? No explanation is given. The observation is recorded in students' science notebooks and the Driving Question Board is opened. This phenomenon anchors all eight lessons: every factor investigated (temperature, concentration, surface area, catalysts, light) is connected back to the question of what must be happening at the particle level inside the beaker to cause the dramatic difference in speed — and how that same principle explains fermentation of uji, rusting of iron roofing sheets, cooking of githeri, and enzyme activity in the human gut.
Storyline Thread	Lesson 1 — What Is a Rate of Reaction? (Anchoring Phenomenon Launch) Students observe the disappearing thiosulfate cross demonstration (cold vs. warm beaker). They record observations, generate initial explanations, and post questions on the Driving Question Board (DQB). The teacher introduces the formal definition of rate of reaction and discusses methods of measuring it (time for precipitate to form, volume of gas collected, loss of mass). Students make initial predictions about what caused the difference in speed between the two beakers and sketch a beginning particle model. Lesson 2 — Measuring Reaction Rates: Designing the Investigation Students learn to quantify reaction rate using the thiosulfate–HCl system (rate = 1/time for cross to disappear). They plan a controlled investigation, identifying independent variable (one factor to change), dependent variable (time/rate), and controlled variables. Groups design a data table and predict the shape of a rate-vs.-factor graph. The DQB is updated with new questions emerging from the planning process. Lesson 3 — Factor 1: Effect of Temperature on Reaction Rate (Experiment) Groups carry out the thiosulfate–HCl experiment at four temperatures (cold water ~10 °C, room temperature ~25 °C, warm ~40 °C, hot ~55 °C). They record time for the cross to disappear, calculate rate (1/time), and plot a graph of rate vs. temperature. Class data is pooled. Students construct an explanation using particle collision theory: higher temperature → particles move faster → more frequent and more energetic collisions → faster reaction. The mango-rotting and ugali-cooking supporting phenomena are revisited.

	Lesson 4 — Factor 2: Effect of Concentration on Reaction Rate (Experiment) Students investigate the effect of varying sodium thiosulfate concentration (keeping temperature and volume constant) on reaction rate. They dilute the thiosulfate to four concentrations, measure time for cross to disappear, and plot rate vs. concentration. Collision theory explanation is extended: higher concentration → more particles per unit volume → more frequent collisions → faster reaction. Students connect this to the antacid tablet supporting phenomenon and to why a stronger acid in a car battery corrodes metal faster. Lesson 5 — Factor 3: Effect of Surface Area on Reaction Rate (Experiment) Using marble chips (calcium carbonate) and dilute hydrochloric acid, groups measure the volume of CO_2 gas produced per minute using large chips, small chips, and powder. They plot volume-vs.-time curves and compare gradients (rates). Collision theory: smaller particles → greater surface area exposed → more collision sites → faster reaction. Students connect this to why maize meal (unga) dissolves into uji faster than whole maize kernels, and why the crushed antacid tablet works faster. Lesson 6 — Factor 4: Catalysts and Light (Demonstration + Research) The teacher demonstrates the liver-and-hydrogen-peroxide experiment (fresh liver, boiled liver, sand). Students observe, explain using the concept of enzymes as biological catalysts, and distinguish catalysts from reactants. The class discusses industrial catalysts (e.g., iron catalyst in the Haber process for fertiliser production, used at Kenya's Athi River). The effect of light on reaction rate is introduced via the photographic film analogy and the role of light in photosynthesis. Students use available digital or print media to research optimum conditions in one Kenyan industrial or biological process and present findings to the class. DQB is revisited and updated. Lesson 7 — Rates of Reactions in Biological, Chemical, and Physical Processes (Application and Sense-making) Students synthesise all four factors into a unified collision-theory model. They apply their model to explain: enzyme kinetics in digestion (optimum temperature/pH), fermentation of uji and yoghurt (temperature control), rusting of iron-sheet roofs in coastal Mombasa (high humidity and salt concentration), and food preservation strategies (refrigeration, salting, drying of fish at Lake Victoria). Groups create annotated diagrams linking each Kenyan context to the relevant factor(s) affecting reaction rate. The DQB is reviewed: which original questions have been answered? Which remain? Lesson 8 — Model Building, Assessment, and Community Connection (Synthesis Lesson) Students revise and finalise their cumulative particle-collision model, incorporating all four factors and both biological and chemical contexts. Groups present their models and explain how manipulating reaction-rate factors can solve a real Kenyan problem of their choice (food spoilage, industrial efficiency, medical dosage, environmental corrosion). A brief written assessment (predict–explain task using a novel Kenyan scenario) is administered. The DQB is closed out: students vote on the most satisfying answer to the driving question and identify remaining curiosities for further study. The lesson closes by connecting to careers (food scientist, industrial chemist, pharmacist, KEMRI biochemist) and to the KICD values of responsibility and respect in scientific work.
Supporting Phenomena	– Supporting Phenomenon 1 — The Rotting Mango Problem: A market trader in Kisumu notices that mangoes left in the open stall at 35 °C go soft and mouldy within two days, while identical mangoes kept in a cool box at 8 °C stay fresh for over a week. Students are asked: is this a chemical/biological reaction? What factor is changing? How does this connect to the disappearing cross? – Supporting Phenomenon 2 — Faster Fizz with Crushed vs. Whole Antacid Tablets: A student at a Nairobi chemist is given an antacid tablet for stomach acid (excess HCl). She notices the crushed powder fizzes and relieves pain far faster than the whole tablet. Why does breaking the tablet into smaller pieces speed up the reaction between the antacid (calcium carbonate) and stomach acid?

	– Supporting Phenomenon 3 — Tea Factory Catalyst Mystery: Workers at a Kericho tea factory add small amounts of a special enzyme (oxidase) to speed up the oxidation of tea leaves during processing. Without the enzyme, oxidation takes hours; with it, the same reaction completes in minutes — yet the enzyme is recovered unchanged at the end. How can something speed up a reaction without being used up? – Supporting Phenomenon 4 — Hydrogen Peroxide and the Liver: A biology teacher demonstrates that adding a small piece of fresh cow's liver (obtained from a Nairobi butchery) to hydrogen peroxide produces a dramatic burst of oxygen bubbles, while adding sand produces nothing. Boiled liver produces almost no bubbles. Students must explain what the liver contains, why heat destroys the effect, and how this connects to reaction rates in the human body.

LESSON 1 (40 minutes): What Is a Rate of Reaction? — Anchoring Phenomenon Launch	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the sub-strand by anchoring all future learning in a single dramatic, puzzling observation — the disappearing thiosulfate cross — and to establish the meaning of 'rate of reaction' as a measurable quantity.
Knowledge	Learners describe the rate of a chemical reaction as a measure of how quickly reactants are converted to products over time, expressed as change in a measurable quantity per unit time.
Skills	Learners observe a chemical reaction carefully, record qualitative and quantitative data in their science notebooks, and formulate initial questions and predictions about what causes differences in reaction speed.
Attitudes	Learners appreciate that chemistry is observable in everyday Kenyan life — from the fast boiling of uji to the slow rusting of iron roofing sheets — and develop curiosity about particle-level explanations for macroscopic changes.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lesson 1 opens the entire sub-strand storyline. The disappearing thiosulfate cross creates a cognitive anchor: two identical reactions, dramatically different speeds. Every subsequent lesson (temperature, concentration, surface area, catalysts, light) will return to this moment and add a layer of particle-level explanation. The Driving Question Board opened today accumulates evidence across all eight lessons.
Safety Notes	Dilute hydrochloric acid (HCl, ≤1 mol/dm³) and sodium thiosulfate solution are used. Learners must NOT lean directly over beakers — sulfur dioxide gas is produced in small amounts and must not be inhaled. Conduct the demonstration in a well-ventilated room or near an open window. If acid contacts skin, rinse immediately with plenty of water and inform the teacher. Safety goggles must be worn by the teacher during setup. Learners observe from a seated distance of at least 60 cm.

B. LESSON OVERVIEW

In this 40-minute opening lesson, the teacher launches the anchoring phenomenon — the Disappearing Thiosulfate Cross — using two beakers of sodium thiosulfate and HCl placed over paper crosses in water baths at contrasting temperatures (≈10 °C simulating a cool Kericho/Nairobi highland morning, and ≈50 °C simulating a hot Mombasa or Kisumu afternoon). Learners first predict what they expect to see, then observe the dramatic difference in how quickly the black cross disappears, then begin to explain by constructing an initial definition of 'rate of reaction'. The lesson closes by formally opening the Driving Question Board (DQB) with learner-generated questions, and each student makes a first sketch of their initial explanatory model in their science notebook. No full explanation is given: the puzzle remains deliberately open, propelling curiosity into Lesson 2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Common	Learners sit in groups of four. Each learner is given a half-page 'Prediction Slip' with two columns:	1. Place apparatus visibly on the demonstration bench before learners enter so curiosity begins	Individual Writing then Pair-Share (Think-Pair-Share variant): By requiring written individual	Teacher scans Prediction Slips as learners write — look for: (a) learners who already mention

polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	(A) What I think will happen and (B) Why I think so. The teacher shows the two beakers, two paper crosses, the cold-water tray labelled 'Kericho Morning — ≈10 °C' and the warm-water tray labelled 'Mombasa Afternoon — ≈50 °C', and announces that the same chemical solutions will be poured into both simultaneously. Learners write individual predictions silently for 3 minutes, then share with their partner for 1 minute. Four pairs are cold-called to share predictions with the class. Learners discover the class is divided — most predict both beakers will behave the same; a few predict the warm beaker will be different. All predictions are recorded without judgment on the chalkboard in two columns: 'Same speed' / 'Different speed'.	immediately. 2. Say exactly: 'Look carefully at what I have set up. Both beakers will receive the same chemicals — same amounts, same concentrations. I want you to predict, in writing, what you think will happen when I pour the sodium thiosulfate into each beaker. You have three minutes — write in your notebook now.' 3. Start a visible 3-minute timer. After 3 minutes say: 'Share your prediction with your partner. You have one minute.' 4. After 1 minute, cold-call exactly FOUR pairs using a random name-stick method. Ask each: 'Tell us your prediction AND your reason.' Allow 10 seconds of WAIT TIME between calling a name and expecting a response. 5. Write all predictions on the board under two columns without saying who is right. Say: 'We are going to find out together. Let us observe carefully.'	predictions before discussion, the teacher activates prior knowledge from everyday Kenyan experience (e.g., why githeri takes longer to cook on a cold highland morning than in the hot lowlands) and surfaces misconceptions (e.g., 'same chemicals always react the same way') that the observation will productively disrupt.	temperature as a variable (these learners have useful prior knowledge to surface); (b) learners who give no reason (these learners need prompting with 'What do you know about temperature and cooking or food spoilage that might help?'). Cold-call responses reveal whether learners connect chemical reactions to everyday phenomena. Indicator of readiness: at least 50% of learners can state a reason, however naive, for their prediction.
Observe Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher performs the demonstration live. Learners watch from their seats, science notebooks open to a fresh page titled 'The Disappearing Cross — [today's date]'. As the thiosulfate is poured simultaneously, a learner volunteer (selected before class) starts a stopwatch for each beaker. The class watches in silence. Within 30–40 seconds, Beaker B (warm/Mombasa) turns milky-yellow and the cross disappears; learners gasp or comment spontaneously. Beaker A (cold/Kericho) remains clearer for 2–4 minutes. Learners record in their notebooks: (1) A time-stamped description of what they see in each beaker at 15 s, 30 s, 60 s, 90 s, and 120 s. (2) The time	1. Before pouring, say: 'I need complete silence and full attention. Open your notebooks. Every 15 seconds I will call out the time — you write what you see in each beaker.' 2. Call out times: '15 seconds… 30 seconds… 45 seconds…' in a calm, measured voice. 3. When Beaker B's cross disappears, say loudly: 'Beaker B — cross gone! Note the time.' Do NOT explain why yet. 4. After both crosses have disappeared, say: 'Take one minute — share with your partner exactly three observations you made. Use the words: I saw… I noticed… I measured…' 5. Cold-call THREE learners: 'What was the most surprising thing you observed?' Allow 10 seconds WAIT TIME	Structured Observation with Time-Stamped Data Recording: By requiring timed, sequential observations rather than a single end-point note, learners build the concept that a reaction has a rate — it happens over time — before the word 'rate' is formally introduced. The contrast between the two beakers makes the variable (temperature) salient without naming it yet, preserving productive inquiry.	Teacher circulates during the pair-share and checks notebooks for: (a) presence of time-stamped entries (observable indicator: at least 3 time points recorded); (b) whether learners describe the milky-yellow precipitate specifically or just write 'it changed colour' (the former shows more careful observation). Ask two or three learners: 'What exactly did you see forming in the beaker?' — look for the word 'cloudy', 'milky', or 'solid forming'. Learners who cannot describe the precipitate need guided re-observation before Phase 3.

	at which each cross disappears. (3) A quick sketch of each beaker at the moment the cross first disappears. After Beaker A's cross also disappears, the teacher asks: 'What did you observe?' — learners share observations in pairs for 1 minute before a whole-class debrief.	each. 6. Record key observations on the board in learner language — do not correct or explain yet. 7. Say: 'We have the SAME chemicals. SAME amounts. But dramatically different speeds. This is our puzzle for the next eight lessons.'		
Explain Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos] 📖 **HTML:** [Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX)] Source: CK-12 [🔍 Search ARES for similar readings]	The teacher writes the word RATE on the board and asks: 'In everyday Kenyan life, when do we talk about how fast something happens?' Groups brainstorm for 90 seconds and share: matatu speed on Thika Road, how fast uji ferments, how quickly maize dries in the Rift Valley sun. The teacher then bridges: 'In chemistry, we also talk about how fast a reaction happens — we call this the rate of reaction.' Learners are guided to write the following working definition in their notebooks: 'The rate of a chemical reaction is a measure of how quickly reactants are used up or products are formed over time.' The teacher then asks: 'Looking at our two beakers — how would you measure the rate of this particular reaction?' Groups discuss for 2 minutes and propose measuring methods (time for cross to disappear, change in cloudiness, change per second). The teacher affirms: 'You have just described how scientists measure rate — by tracking a measurable change over time.' Learners write a personal sentence: 'In our experiment, I could measure the rate of reaction by…'	1. Write: RATE on the board. Ask: 'In Kenya, when do we describe how fast something happens in daily life? Think of examples from your home, your kitchen, your journey to school. You have 90 seconds in your group.' 2. Cold-call FOUR groups — accept all answers, bridge each to chemistry: 'Yes — a matatu travelling at 80 km/h on the Nairobi–Thika highway — that is a rate of distance change. In chemistry, we measure a rate of chemical change.' 3. Write the definition on the board; learners copy it verbatim into their notebooks. 4. Ask: 'How exactly could you measure the rate of THIS reaction — the one you just observed? What could you count or time?' Allow 10 seconds WAIT TIME. Cold-call THREE learners. 5. Affirm and extend each response. 6. Say: 'In your notebook, complete this sentence: In our experiment, I could measure the rate of reaction by… You have 90 seconds.' 7. Cold-call TWO learners to share their sentences. Correct any misconceptions gently: 'Remember — rate involves change AND time. Can you add the time part to your sentence?'	Concept-Bridging from Everyday Context to Scientific Definition: By anchoring the abstract term 'rate' in familiar Kenyan phenomena (matatu speed, fermentation of uji, drying of maize) before applying it to the chemical experiment, learners construct personal meaning rather than memorising a definition. Writing a personal sentence consolidates individual understanding and reveals gaps for the teacher.	Collect 5–6 'personal sentence' responses during cold-call. Indicator of success: the sentence includes both a measurable change AND a reference to time (e.g., 'I could measure the rate by timing how long it takes for the cross to disappear'). Sentences that mention only the endpoint ('when the cross is gone') — without time — indicate the learner has not yet grasped rate as a time-dependent quantity. Teacher notes these learners for targeted support in Lesson 2.
Driving Question Board (DQB)	Each learner receives one sticky note (or tears a small piece of paper). The teacher says: 'Our big	1. Distribute sticky notes. Say: 'Think about what you observed today. Write one question —	Driving Question Board (DQB) as a Living Knowledge Tracker: The DQB externalises learner curiosity	Quality of DQB questions reveals depth of engagement with the phenomenon. Indicator of high

Creation 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	driving question for this whole topic is: WHY do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive?' This is written on a large piece of manila paper pinned to the wall as the DQB header. Learners individually write ONE question that the experiment raised for them — something they do not yet know the answer to. Questions are collected and the teacher reads 6–8 aloud, grouping similar questions by sticking them in clusters on the DQB under emerging headings (Temperature?, Amount of chemicals?, Type of chemicals?, Something inside the particles?). The teacher says: 'These are YOUR questions. Over the next seven lessons, we will answer every cluster on this board — and every time we answer one, we will come back here and mark it.'	something you genuinely want to know — that this experiment left unanswered for you. One question per note. You have 90 seconds.' Start timer. 2. Circulate and nudge learners who are stuck: 'What surprised you most? Ask a question about that.' 3. Collect notes. Read 6–8 aloud slowly. As you read each, ask the class: 'Who else had a similar question?' — show of hands. Cluster similar questions together on the DQB. 4. Name the clusters as they emerge: write a tentative heading above each cluster in a different colour marker. 5. Say: 'This board belongs to you. We will return to it at the start and end of every lesson. At the end of Lesson 8, I expect every cluster to have an answer — written by you.' 6. Take a photograph of the DQB (if a phone/camera is available) for the ARES platform record.	and makes the intellectual journey visible. By clustering questions, the teacher helps learners see that their individual puzzlement is shared and structured — this builds a sense of collaborative inquiry. Returning to the DQB in every lesson reinforces that science is an iterative process of asking and answering questions, not a fixed body of facts to memorise.	engagement: questions that reference the particle level or a specific variable (e.g., 'Does the heat make the tiny particles move faster?', 'What if we used more thiosulfate?'). Indicator of surface engagement: questions that restate the observation ('Why did Beaker B go cloudy?'). Teacher uses question quality to gauge readiness for Lesson 2's deeper investigation and to identify which learners need additional scaffolding through real-life analogies.
Model Building Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 4 minutes, learners open their notebooks to a fresh page titled 'My Explanatory Model — Lesson 1'. They draw a simple sketch of the two beakers (cold and warm) and, inside each, draw what they IMAGINE might be happening to the tiny particles of the chemicals. No right answer is expected — this is an initial model. The teacher says: 'Draw whatever you think is happening inside the liquid at the particle level. Use dots or circles to represent particles. Add arrows if you think something is moving. Label your drawing: this is my Lesson 1 model — it will change.' Learners have 3 minutes to draw silently. In the last minute, two volunteers share their models	1. Say: 'Before we finish, I want you to do something brave — draw what you IMAGINE is happening inside these liquids at the particle level. Not what you know — what you imagine. You have three minutes. Start now.' 2. Circulate quietly during drawing — do NOT correct models; make encouraging comments: 'Interesting idea — I wonder what would happen to those particles if we heated them.' 3. After 3 minutes, cold-call TWO learners to share. Say: 'Tell us what your drawing shows.' Allow each learner 20 seconds to explain. 4. After both share, say: 'Notice how different our two models are — that is perfectly normal at the start of an	Initial Model Drawing (Anchored Particle Thinking): Asking learners to draw particle-level models before instruction is given serves two purposes: (1) it reveals existing mental models of matter (particles vs. continuous substance) that the teacher can build on or address; (2) it establishes the habit of particle-level thinking that is the explanatory thread running through all eight lessons. The explicit statement that 'this model will change' normalises revision as a sign of learning, reducing anxiety about being wrong.	Teacher photographs or mentally notes the range of initial models. Key indicator: do learners draw discrete particles (circles/dots) or a continuous liquid with no internal structure? Learners who draw no particles at all may need a brief review of particle theory at the start of Lesson 2. Learners who spontaneously draw faster-moving particles in the warm beaker already have a productive intuition that will be formally developed in Lesson 2 (temperature and collision theory).

	under the document camera or by holding them up. The teacher acknowledges both without correcting: 'Excellent starting points. We will revise these models in every lesson as we gather more evidence.'	investigation. A model is a tool we improve with evidence. By Lesson 8, your model will look very different from today's — and that means you have learned.' 5. Close by restating the driving question: 'Why do some reactions happen in a flash while others take days? We do not fully know yet — but we have started. See you in Lesson 2.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your teaching journal:

1. PHENOMENON IMPACT — Did the observation of the two beakers produce genuine surprise and curiosity? If learners were not puzzled, consider whether the temperature contrast was sharp enough (check that the warm bath was actually at ≈50 °C) or whether the demonstration bench was visible to all learners. Reposition desks for Lesson 2 if needed.

2. DQB QUALITY — Read through all the sticky-note questions collected today. Are there questions that point toward temperature, concentration, surface area, catalysts, or light? These are the variables you will investigate in Lessons 2–6. If key variables are missing, you may need to plant a seed question yourself in Lesson 2 ('I noticed nobody asked about…').

3. PARTICLE MODELS — How many learners drew discrete particles vs. a continuous liquid? If more than one-third drew no particles, plan a 5-minute particle-theory review warm-up at the start of Lesson 2 using the analogy of githeri grains (discrete) vs. uji porridge (appears continuous but is made of particles).

4. EQUITY CHECK — Were learners at the back of the room able to observe the beakers clearly? For Lesson 2, consider using a mirror or phone camera projected on a wall to give all learners an equal view of the demonstration bench.

5. TIMING — Did the warm beaker's cross disappear within 40 seconds? If the reaction was slower, the sodium thiosulfate may need to be freshly prepared or the concentration increased slightly. Record the actual concentrations used today for reproducibility in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two beakers containing identical amounts of sodium thiosulfate solution and dilute hydrochloric acid were placed over paper crosses. Beaker A was in cold water (≈10 °C — like a cool Kericho morning) and Beaker B was in warm water (≈50 °C — like a hot Mombasa afternoon). A milky-yellow solid (sulfur) formed in both beakers, but the cross under Beaker B disappeared in approximately 30–40 seconds, while the cross under Beaker A remained visible for several minutes.
What did I learn?	The rate of a chemical reaction is a measure of how quickly reactants are used up or products are formed over time. In this experiment, the rate could be measured by timing how long it took for the cross to disappear. The two beakers had the same chemicals in the same amounts, yet their rates of reaction were dramatically different — something about the conditions of the reaction was changing the speed.
How does this explain the phenomenon?	We cannot yet fully explain WHY the warm beaker reacted faster. Our initial particle models suggest that something at the particle level inside the warm beaker is different — perhaps the particles are moving differently or interacting more frequently. We have opened our Driving Question Board with questions about temperature, particle movement, and other possible factors. These questions will be investigated and answered across Lessons 2–8.

LESSON 2 (80 minutes (2 × 40-minute lessons)): What Is Rate of Reaction? — Measuring the Speed of Change	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To formally define rate of reaction, identify measurable quantities that track reaction speed, and connect those measurements to the anchoring phenomenon of the disappearing thiosulfate cross observed in Lesson 1.
Knowledge	Learners describe rate of chemical reaction as the change in amount of reactant consumed or product formed per unit time. They identify three measurable methods: (1) time taken for a precipitate to obscure a reference mark, (2) volume of gas collected over time, and (3) loss of mass over time. They state that a faster rate means shorter time for the same observable change.
Skills	Learners perform a timed precipitate-formation experiment (thiosulfate cross method), record quantitative time data, calculate a simple rate (1/time), plot results, and sketch an initial particle-level model explaining why Beaker B (warm) reacted faster than Beaker A (cold).
Attitudes	Learners appreciate that careful, repeatable measurement is the foundation of scientific knowledge — and that controlling variables (same concentrations, same volumes) is what makes comparisons valid and trustworthy.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lesson 1 established the puzzling observation: two identical beakers of chemicals reacted at very different speeds. Lesson 2 gives learners the language and measurement tools to describe that difference precisely — 'rate of reaction' — and opens the question of WHY temperature changed the rate, which drives the remaining six lessons of the sequence.
Safety Notes	Dilute hydrochloric acid (HCl) and sodium thiosulfate solution are used. Learners must wear safety goggles and avoid skin contact. Sulfur dioxide gas is produced in small quantities — ensure adequate ventilation or conduct the demonstration near an open window. Broken glassware must be reported immediately to the teacher. No eating or drinking in the laboratory. Wash hands thoroughly after the activity. First-aid kit must be accessible.

B. LESSON OVERVIEW

Learners open by revisiting their Lesson 1 science notebook observations of the cold vs. warm thiosulfate beakers and articulate what they still do not understand. The teacher formally introduces the concept of rate of reaction and three standard methods of measuring it. Learners then conduct a structured repeat of the cross-disappearance experiment at room temperature only, practising precise timing and data recording. They calculate a simple rate value (1/t) for each trial, compare results across groups, and discuss sources of variation. The teacher models connecting the new vocabulary back to the anchoring phenomenon. Learners sketch an initial particle model in their notebooks — showing particles of sodium thiosulfate and HCl — and post new questions on the Driving Question Board to guide subsequent investigations into temperature, concentration, surface area, catalysts, and light.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Common	Learners open their science notebooks to yesterday's Lesson 1 observations. Working in pairs,	Display both beakers on the demonstration bench (cold water tray still visible from Lesson 1).	Think-Pair-Share with Notebook Anchor: learners retrieve prior observational data before	Listen for: (a) whether learners distinguish between speed of reaction and amount of product

polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	they re-read their recorded descriptions of Beaker A (cold, ~10 °C, simulating a cool Kericho highland morning) and Beaker B (warm, ~50 °C, simulating afternoon heat in Mombasa). Each pair writes two sentences answering: 'How would you explain the difference in speed to a younger sibling in Standard 6?' They also predict: 'If we repeated the warm-beaker experiment three times, would we always get exactly the same time for the cross to disappear? Why or why not?' Pairs share predictions with the class before any new instruction is given.	Say: 'Before I tell you anything new today, I want you to look back at what YOU observed yesterday. Open your notebooks to your cross-disappearance sketches.' Allow 2 minutes of silent individual review. Then say: 'Turn to your neighbour — you have 90 seconds. Explain the speed difference in words a Standard 6 pupil would understand. Go.' After pairs talk, cold-call 3 pairs using name cards: 'Amina, what did your pair decide?' — 'Kipchoge, do you agree or disagree with Amina — and why?' — 'Wanjiku, your pair had a different idea. Share it.' Apply 10 seconds of WAIT TIME after each cold-call before accepting the answer. Record key student phrases ('the hot one reacted faster', 'the warm one made more sulfur quickly') on the board under the heading 'OUR IDEAS SO FAR'. Do NOT correct or validate yet — affirm participation: 'Thank you — we are going to test all of these ideas over the next few lessons.'	receiving new instruction, activating episodic memory of the phenomenon and surfacing existing mental models. By articulating explanations in everyday Swahili-inflected language ('pole pole' vs 'haraka'), learners identify the conceptual gap that the lesson will fill — the need for a precise, measurable definition of 'speed' of a reaction.	(observable indicator: they say 'faster' or 'took less time', not just 'more sulfur'); (b) whether learners already use the word 'temperature' as a cause (flag these students — they will serve as peer explainers later). If fewer than 3 pairs mention time as the key variable, prompt: 'How did you know one was faster? What did you actually measure or count?'
Observe Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES	Groups of 4 conduct the cross-disappearance experiment at room temperature (approximately 25 °C, representing a typical Nairobi classroom). Each group receives: a 250 mL conical flask, a sheet of white paper with a bold black cross (×) drawn in permanent marker, 50 mL of 0.15 mol/L sodium thiosulfate solution, 5 mL of 1.0 mol/L dilute HCl, a stopwatch, and a thermometer. Procedure: (1) Place the flask over the paper cross. (2) Measure and record the temperature of the sodium thiosulfate solution. (3) Pour the HCl into the flask and immediately start the stopwatch. (4) View the	Before groups begin, deliver a 5-minute explicit instruction segment: write on the board — 'RATE OF REACTION = change in measurable quantity ÷ time taken' and 'A FASTER REACTION = SHORTER TIME for the same change.' Say: 'Scientists need numbers, not just words like fast or slow. Today we are going to measure the rate of our thiosulfate reaction precisely — just like a pharmacist in Nairobi measuring the exact concentration of a drug.' Demonstrate the correct viewing angle (from directly above the flask, not from the side) using the teacher demonstration setup.	Quantitative Data Collection with Class Aggregation: by pooling results from all groups into a single class table, learners immediately see natural variation in room-temperature results (different groups, same conditions). This introduces the concept of experimental error and the importance of repeats — a core NGSS Science and Engineering Practice (Planning and Carrying Out Investigations; Analysing and Interpreting Data). The class table also creates a shared data set that all subsequent discussions can reference, mirroring how Kenyan scientists at KEBS or KEMRI	Circulate and check: (a) Is each group recording time in seconds (not minutes:seconds)? (b) Are they computing 1/t correctly (observable: calculator used, answer expressed in s^{-1} with correct decimal places)? (c) Does the recorded equation show S(s) as the precipitate? Red flag: any group whose three times differ by more than 20 seconds — intervene to coach consistent technique. Green flag: groups that spontaneously note 'our room temperature is 24 °C — that's between cold and warm, so our rate should be between Beaker A and Beaker B from yesterday.'

for similar readings	cross from directly above. (5) Stop the stopwatch the moment the cross is no longer visible. Record the time in seconds. Groups repeat the experiment three times, recording all three times and computing the mean. They then calculate rate = 1/t (in s^{-1}) for each trial and for the mean. Finally, each group records the equation for the reaction: $Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + SO_2(g) + S(s)$, noting that the sulfur precipitate (S) is what clouds the solution and obscures the cross.	Warn: 'Different viewing angles will give you different results — this is a source of error. Each group must agree on ONE person who looks from directly above, every single time.' Circulate during the experiment. At each group, ask: 'What variable are you keeping the same between your three trials? Tell me two things.' Apply 10-second WAIT TIME. Cold-call at least 4 different groups. If a group's three times differ by more than 15 seconds, prompt: 'Your times are quite spread out — what might you change in Trial 3 to get more consistent results?' After all groups record data, display a shared class results table on the board (or a large sheet of manila paper) and have one representative from each group write their mean time and mean rate (1/t).	aggregate trial data before drawing conclusions.	Highlight this publicly as excellent scientific thinking.
Explain Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class discussion connecting today's quantitative data to the Lesson 1 phenomenon. Learners are guided to: (1) Describe, using the formal definition, what rate of reaction means for the thiosulfate experiment. (2) Compare the three methods of measuring rate (precipitate timing, gas volume, loss of mass) using a structured comparison table in their notebooks — columns: Method | What you measure | Suitable reaction example in Kenya. For the third column, learners brainstorm Kenyan examples: gas collection could be used for the fizzing of Eno antacid salts (carbon dioxide); loss of mass could track the burning of charcoal (mkaa) used in a jiko. (3) Sketch an initial particle model in their notebooks: two	Open the Explain phase by pointing to the class data table: 'Look at our class data. Group 3 got a mean time of 48 seconds; Group 7 got 61 seconds. Both worked at room temperature. Does this mean rate of reaction is useless as a measurement? Think for 15 seconds.' (Allow 15 seconds of silence — enforce this visibly by looking at the clock.) Cold-call 2 learners: 'Ochieng, what do you think?' then 'Fatuma, do you agree?' Guide the discussion toward the idea that variation is normal and repeats + means reduce error — do not dismiss variation as 'wrong.' Then explicitly introduce and write the formal definition: 'RATE OF REACTION: the change in concentration of a reactant or product per unit time — or, in our experiment, 1 divided by	Structured Comparison Table + Particle Model Sketch: the comparison table builds disciplinary literacy by organising three measurement methods into a reusable framework learners can apply to any reaction throughout the unit. The particle model sketch is a generative sensemaking tool (NGSS Practice: Developing and Using Models) — by drawing particles rather than just writing words, learners externalise their mental model and make implicit assumptions visible for peer and teacher review. Connecting the model to Kenyan contexts (charcoal jiko, Eno antacid) grounds abstract chemistry in lived experience.	Collect and scan (or circulate to read) three groups' comparison tables. Check: (a) Does the 'gas volume' row identify a realistic Kenyan reaction? (b) Does the 'loss of mass' row correctly identify what decreases (a reactant, not a product)? For particle model sketches, look for: particles in Box B are drawn more numerous OR faster (arrows). Common misconception to flag publicly (without naming individuals): 'Some of you drew MORE particles in the warm beaker — but we used the same volume and concentration! The number of particles is the same. What might be different?' This sets up the collision theory discussions of Lesson 3.

	boxes side by side — Box A (cold beaker, few particles, slow movement) and Box B (warm beaker, many particles shown moving fast with motion arrows). Below the sketch, they write one sentence: 'In the warm beaker, the particles of sodium thiosulfate and HCl are [___] so they form sulfur [___].' Learners fill in the blanks individually, then share with their group.	the time for the cross to disappear.' Link back: 'In Lesson 1, Beaker B took about 35 seconds and Beaker A took about 4 minutes. Use our formula — what was the rate in each beaker?' (Allow 2 minutes of calculation.) Cold-call 3 learners to share values. Then introduce the comparison table, model the first row ('Precipitate timing — we measure time in seconds — thiosulfate cross experiment'), and ask groups to complete rows 2 and 3 using prior knowledge of gas reactions and combustion. After 4 minutes, cold-call representatives from two groups to share their Kenyan examples for each method. Respond: 'Excellent — a Nairobi pharmacist measuring how fast an effervescent Aspirin tablet dissolves is doing exactly the same thing as us measuring how fast sulfur forms.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the first (yellow), they write one thing they now understand better than they did at the start of today's lesson — beginning with 'I now understand that...' On the second (pink), they write one question they still cannot answer — beginning with 'I still wonder...' Learners walk to the Driving Question Board (a large sheet of manila paper displayed at the front of class, divided into columns: Temperature | Concentration | Surface Area | Catalysts | Light | Other) and place their yellow note under the column most relevant to their new understanding and their pink note under the column matching their lingering question. The teacher reads aloud five selected pink	Distribute sticky notes and say: 'You have 3 minutes — silent, individual. Write your yellow note first, then your pink note. No conferring yet.' After 3 minutes, say: 'Walk up, place your notes, and return to your seat. You have 90 seconds.' Once notes are placed, read aloud five pink notes selected for their representativeness (e.g., 'Why does heat make it faster?', 'Would it matter if we used more HCl?', 'Does light affect reactions like it does in photography?'). Say: 'These are exactly the questions professional chemists ask. The NCBA chemists who test water quality at Lake Victoria ask the same questions when they need reactions to go faster or slower.' Conduct the vote by show of	Two-Colour Sticky Note DQB Protocol: separating 'what I now understand' (yellow) from 'what I still wonder' (pink) makes metacognitive activity visible and public. Grouping pink notes by factor column (Temperature, Concentration, etc.) previews the lesson sequence for learners without the teacher announcing it — learners feel ownership of the investigation agenda. The class vote builds a community of inquiry, a key CBC value (citizenship and collaboration).	After the session, photograph the DQB and count: (a) How many yellow notes correctly reference 'rate' or '1/time' as a concept? (b) Are any pink notes asking questions that reveal persistent misconceptions (e.g., 'Does the warm beaker have more chemicals?')? Use these data to plan the opening of Lesson 3. If fewer than 60% of yellow notes reference the formal rate definition, plan a 5-minute retrieval warm-up at the start of Lesson 3.

	questions, grouping similar ones. The class votes (show of hands) on which TWO questions feel most urgent to investigate next. These are circled on the DQB and remain visible for Lesson 3.	hands. Circle the two winning questions in red marker on the DQB. Say: 'These two questions are our job for the next lessons. Keep them in mind.' Do NOT answer the questions — preserve productive uncertainty.		
Model Building Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their particle model sketches from the Explain phase and revise them based on the DQB discussion. They add: (a) labels showing '$Na_2S_2O_3$ particles' and 'HCl particles' in each box; (b) a caption beneath each box giving the approximate rate value calculated from the Lesson 1 phenomenon (Beaker A rate ≈ 1/240 s^{-1}; Beaker B rate ≈ 1/35 s^{-1}); (c) a double-headed arrow between Box A and Box B labelled 'What changed? → TEMPERATURE'. At the bottom of the model, learners write a 'Model Claim': 'My model currently explains ___. My model does NOT yet explain ___.' They share their claim with one neighbour and identify one feature of their model they would change after hearing their neighbour's version. Revised models are kept in their science notebooks for updating in every subsequent lesson.	Say: 'A scientific model is never finished — it gets better each time we gather new evidence. Today you are building Version 1 of your particle model. By Lesson 8, your Version 8 model should explain everything on our Driving Question Board.' Demonstrate on the board how to add particle labels and rate captions to the model skeleton drawn earlier. After 5 minutes of individual revision, say: 'Turn to your neighbour. Compare your models — find ONE thing your partner drew that you did not, and add it to your own model with a different coloured pen. You have 2 minutes.' Cold-call 2 pairs to share what they added. Close the lesson by pointing to the DQB: 'Our model today only explains ONE factor — temperature. Count the columns on the board. How many factors are still unexplained?' (Learners count: Concentration, Surface Area, Catalysts, Light, Other.) 'That is exactly what we will investigate in Lessons 3 through 7. Every time we finish a lesson, your model grows. By the end, your model will be powerful enough to explain why githeri cooks faster in a pressure cooker, why uji ferments overnight in a warm kitchen, and why iron roofing sheets rust slower in Nairobi highlands than on the Mombasa coast.'	Cumulative Model Revision with 'Model Claim' Statement: requiring learners to explicitly state what their model explains AND does not explain operationalises the NGSS Practice of Developing and Using Models at a metacognitive level. The 'different coloured pen for additions' technique makes model revision visible and traceable over eight lessons, turning the notebook into a longitudinal scientific record. Linking the model's limitations directly to the DQB columns previews the inquiry sequence and sustains curiosity.	Scan five notebooks (select a stratified sample: high, mid, and lower-confidence learners identified from Lesson 1 participation). Check: (a) Are rate values (1/t) correctly calculated and labelled on the model? (b) Is the 'Model Claim' statement specific (e.g., 'My model explains why warm water speeds up the reaction but does not yet explain why more concentrated HCl might also change the rate') rather than vague? (c) Is the double-headed arrow labelled 'TEMPERATURE' present? Use findings to form targeted groups for the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did the formal definition of rate of reaction (change in quantity per unit time; 1/t) land clearly, or did learners conflate 'rate' with 'amount of product'? Review the DQB yellow notes for evidence. (2) Were the three measurement methods (precipitate, gas, mass loss) linked convincingly to Kenyan real-life contexts, or did they feel abstract? If the charcoal jiko and Eno antacid examples did not resonate, consider sourcing a short video clip of a Kenyan jiko fire for the next lesson's warm-up. (3) Which groups showed the widest variation in their three repeat times? Visit those groups first in Lesson 3 to reinforce technique before introducing the temperature variable. (4) Were learners' particle models at the end of the lesson showing particles with motion arrows (kinetic understanding) or simply 'more particles' in the warm box (quantity misconception)? This distinction is critical for Lesson 3's collision theory introduction. (5) Did the DQB vote surface any unexpected student questions — for example, about light or pressure — that might need to be acknowledged sooner in the sequence than planned?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners repeated the sodium thiosulfate and HCl cross-disappearance experiment at room temperature (~25 °C, typical Nairobi classroom conditions), recording three timed trials per group and pooling class results into a shared data table. They observed natural variation between groups' times and discussed sources of experimental error.
What did I learn?	Learners formally defined rate of reaction as the change in a measurable quantity per unit time, and expressed it as 1/t (s^{-1}) for the cross-disappearance method. They compared three standard measurement methods (precipitate timing, gas volume collection, loss of mass) and connected each to a Kenyan real-life example (thiosulfate cross, Eno antacid fizz, burning mkaa on a jiko).
How does this explain the phenomenon?	Learners constructed an initial particle model distinguishing Beaker A (cold, slow particles, slow rate) from Beaker B (warm, fast-moving particles, fast rate), labelling the rate values calculated from Lesson 1 data. They posted new questions on the Driving Question Board identifying temperature as the first confirmed variable — while noting that concentration, surface area, catalysts, and light remain unexplained, to be investigated in Lessons 3–7.

LESSON 3 (80 minutes): Concentration and Collision Frequency: Why Does a More Concentrated Acid React Faster?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate how concentration affects the rate of a chemical reaction and explain the effect using collision theory at the particle level.
Knowledge	Learners will be able to: (a) describe how increasing the concentration of a reactant increases the rate of reaction; (b) explain the effect of concentration on reaction rate using the collision theory (more particles per unit volume → more frequent effective collisions → faster reaction); (c) define and use the term 'effective collision'; (d) connect concentration effects to the anchoring phenomenon (thiosulfate cross disappearance) and to real-life Kenyan contexts (acid rain on iron roofing sheets, concentration of digestive acids in the gut, preservation of uji by dilution).
Skills	Learners will be able to: (a) design and carry out a fair-test experiment varying concentration while controlling temperature and surface area; (b) measure and record time for the cross to disappear at four different concentrations of sodium thiosulfate; (c) calculate rate of reaction as 1/time and construct a rate vs. concentration graph; (d) identify variables (independent, dependent, controlled) and justify control choices; (e) draw particle diagrams showing low- and high-concentration systems to illustrate collision frequency.
Attitudes	Learners will: (a) appreciate the importance of precise measurement and fair-test design in generating reliable data; (b) show responsibility in safe handling of dilute hydrochloric acid and sodium thiosulfate; (c) value collaborative evidence gathering as the foundation of scientific knowledge; (d) connect chemical knowledge to making life in Kenya safer and more productive (e.g., understanding how concentration of preservatives or acids affects food safety and industrial processes).
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	In Lesson 1 learners observed that temperature dramatically changed how fast the thiosulfate cross disappeared. In Lesson 2 they designed and refined a method for measuring reaction rate (1/time) and identified variables. Now in Lesson 3 they hold temperature constant and vary only concentration, gathering the second piece of evidence needed to answer the driving question. By the end of this lesson learners can explain both the temperature effect (Lesson 1) and the concentration effect (Lesson 3) using the same collision theory framework — a major step toward building the complete particle model on the Driving Question Board.
Safety Notes	Dilute hydrochloric acid (0.5–2.0 mol/dm³) is an irritant — learners must wear safety goggles and avoid skin contact; wash immediately with water if spillage occurs. Sodium thiosulfate solution produces sulfur dioxide gas as a by-product — conduct the experiment near an open window or outside; learners with respiratory conditions should be seated upwind. Do not dispose of thiosulfate/HCl mixtures down the drain without diluting with plenty of water. Broken glassware must be reported and disposed of in a designated sharps container, not a general bin.

B. LESSON OVERVIEW

Learners investigate how changing the concentration of sodium thiosulfate solution affects the rate at which it reacts with dilute hydrochloric acid, using the disappearing-cross method established in Lesson 1. Working in groups of four, they prepare four dilutions of sodium thiosulfate (100%, 75%, 50%, and 25% of stock), time how long the cross takes to disappear at each concentration (temperature and HCl concentration held constant), calculate rate as 1/time, and plot a rate-vs-concentration graph. A guided particle-diagram activity then connects the data to collision theory: more solute particles per unit volume → more frequent effective collisions → faster rate. Learners update their Driving Question Board and revise their cumulative particle model to include a concentration arrow. Kenyan contexts (stomach acid concentration and digestion speed, concentration of preservatives in fermented uji, acid rain intensity on iron roofing sheets in Kisumu) are woven throughout to anchor understanding in real life.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Equivalent Rates (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two images on the board: (A) a cup of strongly brewed Kenyan chai (dark, concentrated) and (B) a cup of very diluted, watery chai. The teacher asks: 'If both cups were at the same temperature, which chai do you think would stain a white cloth faster — and why?' Learners write a silent individual prediction in their science notebooks (2 minutes), answering: (1) Which concentration reacts / stains faster? (2) What is your reason? (3) What do you think is happening at the particle level? They then share predictions with a partner (think-pair) before three cold-call pairs share with the class. The teacher records all predictions verbatim on the Prediction Column of the Driving Question Board without correcting any — including misconceptions.	BEFORE class: prepare two printed images of chai cups (or draw on board) labelled A (concentrated) and B (diluted). Display the prompt: 'Which chai stains a cloth faster — A or B — and what is happening at the particle level?' Step 1 (0:00–0:02): Say exactly: 'Look at these two cups of chai. Both are the same temperature. Write your prediction silently — which one stains a white cloth faster, and WHY? You have 2 minutes — no talking yet.' Step 2 (0:02–0:04): Allow 2 minutes of silent writing. Circulate — do NOT comment on what you see. Observe for misconceptions to address later. Step 3 (0:04–0:06): Say: 'Turn to your partner. Share your prediction and your particle-level reason. You have 90 seconds.' [WAIT — 90 seconds.] Step 4 (0:06–0:09): Cold-call exactly 3 pairs (choose one who predicted 'concentrated is faster', one who predicted 'same speed', one unsure). For each pair ask: 'What is your prediction AND what did you say is happening with the particles?' [WAIT 10 seconds after each question before accepting answer.]	Think-Pair-Share with Prediction Recording. This activates prior knowledge from Lessons 1 and 2, surfaces particle-level thinking (or its absence), and creates cognitive dissonance that the experiment will resolve. Recording wrong predictions publicly normalises scientific uncertainty and gives the teacher real-time formative data on where collision-theory understanding currently sits.	Teacher circulates during silent writing and reads 4–5 notebooks. Observable indicators: (a) Does the learner predict 'concentrated reacts faster'? (b) Does the learner give a particle-level reason (e.g., 'more particles', 'more collisions') or only a macro-level reason (e.g., 'stronger')? Learners who give only macro reasons need scaffolding in Phase 3. Note these learners by name for targeted questioning during the Explain phase.

		Step 5 (0:09–0:10): Record all three predictions verbatim on the DQB Prediction column. Say: 'We are not saying who is right yet — our experiment will tell us. Keep your notebook open on this page.'		
Observe Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Equivalent Rates (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive a pre-prepared set of four labelled conical flasks of sodium thiosulfate solution at different concentrations (prepared by diluting stock with distilled water): Flask 1 = 40 cm^3 stock + 0 cm^3 water (100%); Flask 2 = 30 cm^3 stock + 10 cm^3 water (75%); Flask 3 = 20 cm^3 stock + 20 cm^3 water (50%); Flask 4 = 10 cm^3 stock + 30 cm^3 water (25%). Each flask is placed over an identical black-cross card. A fixed volume of dilute HCl (5 cm^3, same concentration) is added to each flask one at a time (timed by stopwatch). One learner times, one watches from above, one records, one manages safety. Learners record time for cross to disappear for all four concentrations, calculate rate = 1/t for each, and enter results in a shared class data table on the board. Each group then plots a rate (y-axis) vs. relative concentration (x-axis) graph by hand on graph paper, draws the line of best fit, and writes one sentence describing the trend.	BEFORE class: prepare dilutions and label flasks; pre-draw a results table on the board with columns: [Group | Conc. % | Time (s) | Rate = 1/t (s^{-1})]. Step 1 (0:10–0:13): Distribute equipment. Say: 'Your job is a FAIR TEST. Before you start, tell me — what are you keeping the SAME in every trial?' [WAIT 10 seconds.] Cold-call 2 groups. Expected answers: temperature, volume of HCl, same cross card, same observer position. Confirm and say: 'Correct — if those change, your data is not trustworthy.' Step 2 (0:13–0:15): Demonstrate the procedure once with Flask 1 in front of the class. Say: 'Start the stopwatch the MOMENT the HCl hits the thiosulfate. Stop it the MOMENT you can no longer see ANY part of the cross. This is the same method we used in Lessons 1 and 2.' Step 3 (0:15–0:35): Groups conduct the experiment. Circulate every 3–4 minutes. Prompt struggling groups: 'Have you written down your controlled variables? Is your eye position the same each time?' If a group finishes early, prompt: 'Calculate 1/t for each row now. What units	Structured Fair-Test Experiment with Class Data Aggregation. By pooling data from all groups on the board, learners see that the trend (rate increases with concentration) is reproducible across different hands and slight variations — building understanding of scientific reliability. Calculating 1/t reinforces the mathematical definition of rate introduced in Lesson 2, directly connecting evidence-gathering lessons.	Observable indicators during circulation: (a) Can the group correctly identify independent variable (concentration), dependent variable (time/rate), and controlled variables? (b) Are learners using the stopwatch correctly — starting at moment of mixing? (c) Is the 1/t calculation correct with appropriate units? (d) Does the graph have labelled axes, correct scale, and a drawn line of best fit? Flag any group that has a flat or negative trend for targeted questioning — they may have varied temperature accidentally (e.g., warm hands on flasks).

		will rate have?' Step 4 (0:35–0:38): Each group sends one member to write their results on the class table on the board. Say: 'Look at the class data. Do different groups get similar results? If one group's result looks very different, what might explain that?' [WAIT 10 seconds. Cold-call 1 group.] Step 5 (0:38–0:40): Say: 'Now plot rate (1/t) on the y-axis against concentration percentage on the x-axis. Draw the best-fit line. In ONE sentence, describe the trend you see.'		
Explain Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Equivalent Rates (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class Claim-Evidence-Reasoning (CER) discussion. Learners first revisit their Phase 1 predictions and check them against their data. They then complete a structured particle diagram task in their notebooks: draw two boxes side by side — Box A (low concentration, few dots representing thiosulfate particles) and Box B (high concentration, many dots). They draw arrows between dots to show 'collisions' and count how many collision pairs are possible in each box. A guided question sequence connects this to the anchoring phenomenon: 'In the original thiosulfate demonstration, Beaker B was hotter — particles moved faster. Today we kept temperature the same but changed concentration. What is similar about the effect on collisions in both cases?' Learners write a 3-sentence explanation using the terms: effective collision, collision frequency, concentration,	Step 1 (0:40–0:43): Say: 'Look at your Phase 1 prediction. Hold up your notebook. Thumbs up if your prediction was supported by your data. Thumbs sideways if you're unsure. Thumbs down if your data surprised you.' Scan the room — note proportion of thumbs down for adjustment. Step 2 (0:43–0:47): Say: 'Draw two boxes in your notebook. Label Box A: LOW concentration. Label Box B: HIGH concentration. Put 4 dots in Box A and 12 dots in Box B — each dot is one thiosulfate particle. Now draw lines between every pair of dots that could collide. Count the collision lines in each box.' [WAIT 2 minutes for drawing.] Cold-call 1 learner: 'How many possible collisions in Box A? In Box B?' Expected: Box A ≈ 6, Box B ≈ 66. Say: 'This is why concentration matters — it is not magic, it is mathematics of proximity.'	Claim-Evidence-Reasoning (CER) Framework combined with Particle Diagram Construction. CER structures the explanation so that learners must explicitly link their data (evidence) to collision theory (reasoning) rather than stating conclusions without justification. Constructing particle diagrams makes the abstract collision theory visible and countable — a key scaffolding step for learners who gave only macro-level predictions in Phase 1.	Observable indicators: (a) Can the learner correctly use 'effective collision' and 'collision frequency' in a written explanation? (b) Does the particle diagram show proportionally more collision pairs in the high-concentration box? (c) Can the learner articulate the mechanistic link between concentration and rate (more particles per unit volume → more frequent collisions → faster reaction) rather than simply stating 'concentrated reacts faster'? (d) Exit check: hold up notebooks — scan 3-sentence explanations for the four required terms. Any learner missing two or more terms needs a brief one-on-one re-explanation before Phase 4.

	and rate of reaction. Two learners read their explanations aloud for peer critique.	Step 3 (0:47–0:52): Pose the connection question aloud and write it on the board: 'In Lesson 1, raising temperature in Beaker B made the cross disappear faster. Today, raising concentration also made the cross disappear faster. WHAT IS THE SAME about what is happening at the particle level in both cases?' [WAIT 15 seconds — full silence.] Cold-call 3 different learners. Guide toward: both increase collision FREQUENCY (and in the case of temperature, also collision ENERGY). Say: 'Write this in your own words as a rule.' Step 4 (0:52–0:57): Say: 'Now write exactly 3 sentences explaining today's result. You MUST use ALL four of these words —' [write on board]: effective collision / collision frequency / concentration / rate of reaction. 'You have 4 minutes — silent writing.' Step 5 (0:57–1:00): Cold-call 2 learners to read their 3-sentence explanation. After each, ask the class: 'Did they use all four terms correctly? Give a thumbs up for each term you heard used correctly.' Provide brief verbal correction if a term is misused — do not move on until all four terms are used correctly by at least one learner.		
Driving Question Board (DQB) Creation 📹 **VIDEO:**	Each group receives two sticky notes: one GREEN (new evidence/answers) and one ORANGE (new questions raised). On the GREEN note they write:	Step 1 (1:00–1:02): Distribute green and orange sticky notes. Say: 'On your GREEN note, write the key finding from today in ONE sentence — the cause and the	Driving Question Board Gallery Walk with Sticky-Note Annotation. The DQB is a public, cumulative record of the class's growing understanding. Distinguishing	Observable indicators: (a) Is the green sticky note mechanistically accurate — does it mention 'collision frequency' or 'more particles'? (b) Is the orange

Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Equivalent Rates (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'We found that increasing concentration increases rate of reaction because more particles are present so collision frequency increases.' On the ORANGE note they write one new question their experiment raised (e.g., 'Is there a concentration so high that rate stops increasing?' or 'Does concentration affect ALL reactions the same way?'). Groups post their notes on the class DQB. The teacher facilitates a 3-minute gallery scan where learners read others' orange questions, then the class votes (show of hands) on the most interesting new question to add to the main DQB column. The teacher explicitly connects today's lesson to the anchoring phenomenon: 'Think about the thiosulfate cross from Day 1. We have now explained TWO reasons why Beaker B might react faster — temperature AND concentration. How many factors do you think there might be in total?'	particle-level reason. On your ORANGE note, write the most interesting new question your experiment made you think of. You have 2 minutes.' Step 2 (1:02–1:05): Groups post notes on the DQB. Say: 'You have 2 minutes — walk around and read at least two other groups' orange notes silently.' Step 3 (1:05–1:08): Say: 'Hands up — which orange question do you find most scientifically interesting and why?' Cold-call 3 learners. [WAIT 10 seconds after posing the question.] Teacher circles the winning question on the DQB. Step 4 (1:08–1:10): Say: 'In Lesson 1 we saw temperature change the speed of reaction. Today we saw concentration change it. Look at the DQB — we are building a list of factors. How many factors do you predict we will find by the end of this unit?' Record predictions on the DQB. Say: 'Keep those predictions — we will check them in Lesson 8.'	green (answers) from orange (new questions) makes the iterative, never-fully-finished nature of science visible. Voting on orange questions builds agency — learners see that their curiosity drives the next investigation.	question scientifically productive (i.e., testable) or vague? Teacher reads all green notes after class and notes any groups still writing only macro-level explanations (e.g., 'concentrated is stronger') for targeted follow-up at the start of Lesson 4.
Model Building Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identification and	Learners return to their cumulative particle model begun in Lesson 1 (a diagram of the inside of a beaker during the thiosulfate reaction). In Lesson 1 they added temperature arrows and particle speed indicators. Today they annotate the same diagram to show concentration: they add more thiosulfate particles to one version of the beaker and label the effect on collision frequency and rate. They also add a new real-life	Step 1 (1:10–1:12): Say: 'Open your science notebook to the particle model you started in Lesson 1. You are going to ADD to it — do not start a new diagram.' Display a prompt on the board: 'Add to your existing beaker diagram: (1) Draw a second beaker next to it with MORE thiosulfate particles. (2) Show that more collisions are happening. (3) Add a label: higher concentration → higher collision frequency →	Cumulative Model Revision (Iterative Modelling). Returning to and annotating the same diagram across all eight lessons makes the growing explanatory power of the particle model tangible. Each lesson adds one new variable-arrow, so by Lesson 8 learners will have a complete, student-constructed mechanistic model. Connecting to Kenyan real-life contexts (githeri digestion, uji fermentation, Kisumu roofing)	Observable indicators: (a) Does the revised model correctly show more particles in the high-concentration beaker (not just bigger particles — a common misconception)? (b) Is the real-life connection arrow labelled with a specific Kenyan context? (c) Does the closing sentence include particle-level language? Teacher collects notebooks from 6 learners (2 high, 2 middle, 2 low performers based on Lesson 2 data) to review

Writing of Equivalent Rates (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	connection arrow on the model linking 'concentration' to at least ONE Kenyan context chosen from a prompt list: (a) stomach acid (HCl) concentration and speed of protein digestion in githeri; (b) concentration of lactic acid in fermented uji and how diluting it slows souring; (c) intensity of acid rain (higher H^+ concentration) and rusting speed of iron roofing sheets in Kisumu during heavy rains. Learners write one sentence below their model: 'To make a reaction faster, I could increase concentration because ___.'	faster rate.' Step 2 (1:12–1:16): Say: 'Now look at this prompt list —' [display the three Kenyan contexts on the board]. 'Pick ONE context. Draw a small connecting arrow from your concentration label to a real-life example. Label it.' [WAIT 3 minutes for silent drawing.] Step 3 (1:16–1:18): Say: 'Write the sentence stem at the bottom of your model: To make a reaction faster, I could increase concentration because ___. Complete it using particle language.' [WAIT 1.5 minutes.] Step 4 (1:18–1:20): Cold-call 2 learners to read their completed sentence. After each, ask: 'Does that sentence use particle-level language? What word tells us WHY it works?' Confirm that 'collision frequency' or equivalent must appear. Close the lesson by saying: 'In Lesson 4 we will investigate a third factor. Before you come to class, think about what happens when you break a large piece of charcoal into smaller pieces before putting it on a jiko — does it light faster or slower? Write your prediction tonight.'	ensures the abstract model is grounded in lived experience and fulfils the driving question's call to make life in Kenya safer, healthier, and more productive.	models in detail before Lesson 4 and identify persistent misconceptions about particle size vs. particle number.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your teacher journal:\n\n1. MISCONCEPTION CHECK — Did any learners draw 'bigger' particles to represent higher concentration rather than 'more' particles? If yes, plan a 2-minute correction at the start of Lesson 4 using a concrete analogy: 'If we add more ugali portions to a small sufuria, do the portions get bigger — or do we just have more of them in the same space?'\n\n2. GRAPH LITERACY — Were learners able to identify the rate-vs-concentration relationship from their graph as linear (or approximately so) within the range tested? Did any groups draw a curve? Discuss in Lesson 4 what a curve might mean beyond the range.\n\n3. EQUITY OBSERVATION — Which learners did not speak during cold-calls today? Ensure those learners are the first to be cold-called in Lesson 4's Predict phase so that every learner contributes to class discourse at least once across the sequence.\n\n4. KENYAN CONTEXT UPTAKE — Which of the three Kenyan contexts (githeri digestion, uji fermentation, Kisumu roofing) did most learners choose for their model? If uji fermentation dominated, use that as the entry point in Lesson 5 (surface area) to maintain continuity of story.\n\n5. DRIVING QUESTION PROGRESS — How many of the DQB green notes correctly mentioned 'collision frequency'? If fewer than 60% did, consider beginning Lesson 4 with a 5-minute re-teach of the particle diagram before introducing the new variable.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	When we added dilute HCl to four flasks of sodium thiosulfate solution at different concentrations (holding temperature constant), the cross under the most concentrated flask disappeared in the shortest time, and the cross under the most dilute flask took the longest. When we calculated rate as 1/time and plotted a graph, the rate increased as concentration increased.
What did I learn?	Increasing the concentration of a reactant increases the rate of reaction. This is because a higher concentration means more reactant particles are present in the same volume, so particles collide more frequently. More frequent effective collisions mean more product forms per second — a faster reaction rate. This is the same collision theory we used to explain temperature in Lesson 1, but today particle SPEED stayed the same while particle NUMBER per unit volume changed.
How does this explain the phenomenon?	The disappearing-cross phenomenon from Lesson 1 can now be partially explained by concentration as well as temperature: if Beaker B had a higher concentration of thiosulfate AND a higher temperature, both factors would independently increase collision frequency and make the cross disappear faster. In Kenyan contexts: a higher concentration of HCl in the stomach speeds up protein digestion in githeri; a more concentrated lactic acid solution in uji ferments (sours) faster; and more acidic (concentrated H^+) rainfall corrodes iron roofing sheets in Kisumu more rapidly. Controlling concentration is therefore a powerful tool for controlling how fast biological, chemical, and industrial processes run.

LESSON 4 (40 minutes): Factor 2: Effect of Concentration on Reaction Rate — More Particles, More Collisions	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students investigate how changing the concentration of sodium thiosulfate affects the rate of its reaction with hydrochloric acid, and explain the results using collision theory.
Knowledge	Higher concentration means more solute particles per unit volume; more particles per unit volume increases collision frequency; increased collision frequency increases the rate of reaction — consistent with the collision theory model built in Lessons 1 to 3.
Skills	Diluting a solution to prepare four known concentrations; measuring and recording time for the cross to disappear; calculating rate as 1 divided by time; plotting a rate-vs-concentration graph; identifying the relationship as directly proportional; constructing a particle-level explanation from experimental evidence.
Attitudes	Precision and honesty in recording data even when results do not perfectly match predictions; appreciation that systematic dilution is a transferable laboratory skill used by Kenyan pharmacists, food scientists, and KEMRI researchers.
Key Inquiry Question	How does varying the concentration of a reactant affect the rate of a chemical reaction?
Purpose in Storyline	Lesson 4 provides the second experimental factor in the evidence-gathering sequence. Students already know temperature changes particle SPEED; today they discover that concentration changes particle NUMBER per unit volume — a distinct mechanism that still explains faster collisions. Both pieces of evidence feed the cumulative particle model that will be completed in Lesson 7 and assessed in Lesson 8.
Safety Notes	Students must wear safety goggles throughout. Dilute HCl (0.1 mol per litre) and sodium thiosulfate solutions are low hazard at these concentrations but must not be ingested. Spills should be wiped immediately with a damp cloth. The teacher should demonstrate correct dilution technique before groups begin. Wash hands at the end of the practical.

B. LESSON OVERVIEW

Students begin by revisiting the anchoring phenomenon — two beakers with identical chemicals but different temperatures — and are challenged to isolate concentration as a new independent variable while holding temperature constant. Working in groups of four, they prepare four dilutions of sodium thiosulfate (100 percent, 75 percent, 50 percent, and 25 percent of the original concentration) by adding measured volumes of distilled water, then react each solution with a fixed volume of dilute HCl over a paper cross. They record the time for the cross to disappear, calculate rate (1 divided by time), and plot rate vs. relative concentration. The class pools data, identifies a directly proportional relationship, and constructs a particle-level explanation: higher concentration means more thiosulfate particles per unit volume, so collisions with HCl particles are more frequent, producing sulfur precipitate faster. Students connect this to why a stronger stomach acid digests food faster, why antacid tablets are more effective in water, and why a more concentrated battery acid in a matatu corrodes metal faster. The Driving Question Board is updated with new evidence, and students sketch an upgraded particle diagram comparing low and high concentration scenarios.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students open their science	Teacher re-displays the anchoring	Think-Pair-Share anchored to the	Teacher scans notebook

📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Solution Concentration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	notebooks to their particle diagrams from Lesson 3. The teacher displays the anchoring phenomenon image (cold vs. warm beaker with the cross underneath) on the board and asks students to recall what changed in that demonstration. The teacher then poses a new scenario: imagine both beakers are now kept at exactly the same temperature — room temperature — but the sodium thiosulfate in Beaker A is very dilute (mostly water, few thiosulfate particles) while Beaker B uses the full-strength solution (many thiosulfate particles). Students write a prediction in their notebooks: which beaker will make the cross disappear faster, and WHY — using particle language. They then share predictions with a partner (Think-Pair) before three volunteers share with the class. The teacher records predictions on the board in two columns: predictions that say FASTER for concentrated and predictions that say NO DIFFERENCE, noting the reasoning offered for each. Students also sketch two quick particle diagrams side by side — one showing a dilute solution and one showing a concentrated solution — and annotate the difference in particle spacing.	phenomenon image and says: In Lesson 3 we changed temperature and saw that warmer particles move faster and collide more often. Today we are going to keep temperature exactly the same for every beaker. What I am going to change is how many thiosulfate particles are dissolved in the water. Before we do the experiment, I want you to predict what will happen using your particle model. Write your prediction now — you have 90 seconds. WAIT 90 SECONDS IN SILENCE. Then say: Now turn to your partner and compare your predictions. Do you agree or disagree? Explain your reasoning using the word COLLISION. WAIT 60 SECONDS. Then cold-call three pairs. For any student who predicts no difference, the teacher should say: That is an interesting hypothesis — let us keep it on the board and let the evidence decide. Do not correct or validate yet. Teacher then says: Before we start the experiment, sketch two small diagrams in your notebook — one for a dilute solution and one for a concentrated solution. Show the particles of sodium thiosulfate as dots. What is different between the two diagrams? WAIT 60 SECONDS then ask one student to share their sketch using the document camera or by describing it aloud.	particle model; sketched predictive diagrams make implicit mental models visible before the experiment begins, allowing the teacher to identify and address misconceptions early.	predictions during the 90-second writing period and notes whether students are using particle language (number of particles, frequency of collisions) or only macroscopic language (stronger, weaker). Students who use only macroscopic language will be targeted with a probing question during the Observe phase: what is actually happening between the particles to cause what you see?
Observe Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum)	Groups of four receive a practical instruction card, safety goggles, a sheet of white paper marked with a bold black cross, four 250 ml beakers labelled C1 to C4, a measuring cylinder, a stopwatch, dilute HCl (0.1 mol per litre, 10 ml per trial), and a stock solution of	Before groups begin, the teacher demonstrates the dilution technique at the front: I am going to measure 30 ml of sodium thiosulfate using this measuring cylinder — read the bottom of the meniscus at eye level. Then I add 10 ml of distilled water. The	Structured data collection with a shared class data table reduces noise and allows students to see the pattern across multiple groups; pooling data models authentic scientific practice and helps students see that the trend is reproducible even when individual	Teacher checks that students are reading the meniscus correctly during dilution (a common error is reading the top of the meniscus). Teacher also checks that students are calculating 1 divided by time correctly — students who invert the fraction (writing time divided by

🔍 Search ARES for similar videos 📖 **HTML:** Solution Concentration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	sodium thiosulfate (0.15 mol per litre). Groups prepare four concentrations by dilution: C1 — 40 ml thiosulfate plus 0 ml water (full strength); C2 — 30 ml thiosulfate plus 10 ml water (75 percent); C3 — 20 ml thiosulfate plus 20 ml water (50 percent); C4 — 10 ml thiosulfate plus 30 ml water (25 percent). Total volume for each trial is 40 ml thiosulfate solution plus 10 ml HCl = 50 ml. One student acts as timekeeper, one adds the HCl, one observes from directly above, and one records data. For each concentration trial (starting from C4, the most dilute, up to C1, the most concentrated), the group places a beaker of the prepared thiosulfate solution over the cross, adds 10 ml HCl, starts the stopwatch, and records the time in seconds for the cross to become invisible when viewed from above. Results are entered into a pre-drawn data table: concentration label, volume of thiosulfate, volume of water, time in seconds, rate (1 divided by time). After all four trials, groups calculate rates and plot rate (y-axis) vs. relative concentration (x-axis, using 25, 50, 75, and 100 as percentage values). Class results are shared on a whiteboard table and each group adds their data points to a class graph drawn on a large sheet of paper at the front.	TOTAL volume is now 40 ml, but I have fewer thiosulfate particles in that 40 ml than I started with. That is what we mean by a more dilute solution. Teacher circulates during the practical and uses targeted questions: Which trial do you predict will be fastest — start with that one or last? (Answer: last — start dilute so cross remains visible longest, avoiding sulfur contamination of the beaker affecting the next trial.) For groups whose C4 time seems very long, say: That is excellent data — a very slow reaction. What does that tell us about the number of collisions happening per second? For groups whose data seems inconsistent, say: Look at your graph shape — does the trend still hold even if one point is slightly off? Why might one trial have a timing error? WAIT TIME: After asking any probing question, the teacher waits a minimum of 10 seconds before redirecting, modelling that thinking takes time. As groups finish, the teacher invites one representative from each group to add their rate values to the class whiteboard table, then asks the class: What pattern do you see in the class data? WAIT 15 SECONDS before taking responses.	timings vary.	1) will be asked: if a slow reaction gives a large time value, should the RATE be a large or small number? This question redirects the reasoning without giving the answer.
Explain Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Students return to their seats and the teacher facilitates a whole-class sensemaking discussion anchored to three questions written on the board: (1) What did you observe about the relationship between concentration and rate? (2) Why — in terms of particles —	Teacher opens the Explain phase by saying: Look at the class graph on the board. In one sentence, describe the shape of this graph — use the words concentration and rate. WAIT 20 SECONDS. Take three responses. Then say: Now here is the harder question —	Contrastive board diagram (low vs. high concentration particle boxes) makes the mechanistic explanation visible and directly mirrors the particle diagrams students sketched in the Predict phase, creating a before-and-after conceptual arc; explicit	Teacher listens for whether students distinguish between particle SPEED (temperature effect) and particle NUMBER (concentration effect) — a key conceptual boundary. Students who conflate the two will be asked: if I cool down a concentrated

📖 **HTML:** [Solution Concentration (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	does higher concentration cause faster reaction? (3) Where does this pattern show up in everyday Kenyan life? Students respond to question 1 by describing the graph trend (as concentration increases, rate increases — approximately proportional). The teacher then uses a particle animation or a hand-drawn board diagram to reinforce question 2: two boxes are drawn side by side, one labelled LOW CONCENTRATION (few thiosulfate dots, few HCl dots, wide spacing) and one labelled HIGH CONCENTRATION (many thiosulfate dots, same HCl dots, close spacing). Students identify that in the high-concentration box, thiosulfate particles and HCl particles are closer together and encounter each other more often. The teacher explicitly connects this to the language used in Lesson 3: In Lesson 3, more temperature meant particles moved FASTER, so they collided more often. Today, temperature stayed the same — particles moved at the SAME speed — but there were MORE of them, so they still collided more often. The mechanism is the same: more frequent effective collisions. For question 3, students discuss in pairs for 60 seconds before sharing: the antacid tablet dissolved in more water is less concentrated — it works more slowly; a matatu battery with stronger acid corrodes terminal clamps faster; a more concentrated bleach solution removes stains from a kanga faster; a more concentrated sugar solution in uji ferments differently. The teacher records Kenyan contexts on the DQB sticky notes	WHY does this shape make sense? I want you to explain it using particles, not just the words more or less. WAIT 15 SECONDS. Draw the two-box particle diagram on the board as students respond, adding dots as they describe the model. If a student says there are more collisions, probe: more collisions between WHICH particles? What are those particles doing when they collide? What product forms? After the particle explanation is built collaboratively, teacher says: In Lesson 3 we said higher temperature makes particles move FASTER. Today particles moved at the SAME speed — temperature was constant. But the reaction was still faster with high concentration. How do we explain that? WAIT 15 SECONDS. Guide students to the idea that particle NUMBER per unit volume is the new variable. Then say: Turn to your partner and think of one example from your daily life in Kenya where concentration affects how fast something happens. You have 60 seconds. WAIT 60 SECONDS. Cold-call four pairs. Write their Kenyan examples on the DQB as new evidence entries.	comparison to the Lesson 3 temperature mechanism reinforces the unified collision theory model.	solution so the particles move slowly again, will the concentration still matter? This thought experiment targets the misconception that all rate effects reduce to particle speed.

	as students offer them.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solution Concentration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive two sticky notes each. On the first note (green) they write one piece of new EVIDENCE collected today that helps answer the driving question: Why do some reactions happen in a flash while others take days? On the second note (yellow) they write one NEW QUESTION that arose during the lesson — something today's evidence does not yet answer. Students post their notes on the DQB in the designated Lesson 4 section. The teacher reads several notes aloud and places them in two clusters: EVIDENCE WE HAVE NOW and QUESTIONS STILL OPEN. The teacher then draws the class attention to the full DQB, which now contains evidence from three experimental lessons (Lessons 2, 3, and 4) and asks: We now have two factors that speed up reactions — temperature and concentration. What other factors might still be missing from our explanation? Students call out ideas (size of particles, adding another chemical, light) and the teacher previews: In Lesson 5 we will investigate a third factor — the size of the reactant particles. We will use a very Kenyan material — marble chips, the same calcium carbonate that is quarried in Kajiado County — to explore how breaking particles into smaller pieces changes the speed of a reaction.	Teacher distributes sticky notes and says: I want you to do two things. First, write on the GREEN note one thing you observed or calculated today that gives us evidence about what controls reaction speed — use the word CONCENTRATION in your sentence. Second, write on the YELLOW note one question you still have — something today did not fully answer. You have 90 seconds. WAIT 90 SECONDS. Then say: Come up and post your notes. Teacher reads five to six notes aloud, paraphrasing: This group found that halving the concentration roughly doubled the time — that is precise quantitative evidence. This group is asking whether the same pattern holds for the acid concentration rather than the thiosulfate concentration — excellent scientific thinking. Teacher then gestures to the whole DQB and says: Look at what we have built over four lessons. Each lesson we add new evidence to our answer. Which part of the driving question are we closest to answering now? WAIT 20 SECONDS. Take two responses. Then preview Lesson 5 using the marble chip and Kajiado quarry reference to maintain Kenyan relevance.	Dual sticky-note protocol separates evidence (what we know) from curiosity (what we still need to find out), training students in the scientific habit of distinguishing claims from questions; reviewing the full DQB builds a visible cumulative narrative across the eight-lesson sequence.	Teacher scans green sticky notes to assess whether students are making claims at the evidence level (rate increased as concentration increased, cross disappeared faster) or only restating the procedure (we added water to the thiosulfate). Students who only describe procedure will be prompted during Lesson 5 warm-up: what did the procedure SHOW us about particles?
Model Building Phase 📹 **VIDEO:** Common polyatomic ions	In the final five minutes, students open their science notebooks to their cumulative particle model — a diagram they have been developing since Lesson 1. Today they add a new labelled panel	Teacher says: Take out your cumulative particle model from Lesson 3. You are going to add a new panel — the concentration panel. Draw two boxes side by side. In the left box, draw a dilute	Cumulative model building across lessons makes conceptual growth tangible and assessable; the explicit LESSON 3 LINK and LESSON 4 LINK labels help students organise the two	Teacher checks that the concentration panel correctly shows more particles (not larger ones) in the concentrated box, and that the two-sentence summary distinguishes particle number from

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solution Concentration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	showing the concentration effect: two side-by-side boxes (low concentration, high concentration) with dots representing thiosulfate particles and triangles representing HCl particles. In the high-concentration box they draw more dot-triangle collision events with arrows labelled EFFECTIVE COLLISION and label the product (sulfur precipitate — S). Under the diagram they write two sentences: one stating what concentration does to particle number per unit volume, and one connecting this to the rate of the thiosulfate-HCl reaction. At the bottom of the page they write: LESSON 3 LINK — temperature changes particle SPEED; LESSON 4 LINK — concentration changes particle NUMBER. Both increase collision frequency. Students who finish early are invited to add a third sentence connecting their model to one Kenyan context from the Explain phase discussion. At the close of the lesson the teacher says: Your particle model is growing. By Lesson 7 it will explain everything from the cooking pot in your home to the industrial reactors at Athi River.	sodium thiosulfate solution — use dots for thiosulfate particles and triangles for HCl particles. In the right box, draw the concentrated solution. Show what is DIFFERENT between the two boxes. Then show me at least one collision in the right box that leads to a product. You have four minutes. WAIT 4 MINUTES, circulating and checking that students are drawing MORE dots (not larger dots) in the concentrated box — a common misconception is that concentration means bigger particles rather than more particles. If a student draws larger dots, crouch beside them and say: The particles of sodium thiosulfate are always the same size — what changes when we concentrate the solution? WAIT 10 SECONDS for the student to self-correct. At the two-minute mark, say: You should also write the two-sentence summary underneath your diagram — what concentration does to particle number, and what that does to the rate. At the close, teacher says: Your model now has two powerful ideas in it. Temperature changes SPEED. Concentration changes NUMBER. In Lesson 5 we will add a third idea — surface area changes the NUMBER OF SITES where collisions can happen. The marble chips from Kajiado are waiting for us.	mechanisms as parallel but distinct contributions to a unified collision theory.	particle speed. A student who writes both temperature and concentration cause faster collisions because particles move more will receive a targeted prompt in the Lesson 5 warm-up review.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did students successfully distinguish the concentration mechanism (more particles per unit volume) from the temperature mechanism (faster-moving particles) — or did many students conflate the two? If conflation was common, plan a brief comparison activity at the start of Lesson 5. (2) Were students able to transfer the collision theory explanation to Kenyan contexts without prompting, or did they need significant scaffolding? If transfer was difficult, add a Kenya-context matching warm-up in Lesson 5. (3) Did all groups obtain a clear directly-proportional trend in their rate-vs-concentration graph, or did outlier data points create confusion? If data quality was poor, discuss sources of experimental error at the start of Lesson 5 as a modelling-of-science opportunity. (4) Are yellow DQB sticky notes generating questions that anticipate surface area and catalysts (Lessons 5 and 6), or are students still asking questions about temperature and concentration? Use the yellow notes to plan targeted bridging questions for Lesson 5. (5) Were any students disengaged during the dilution preparation step? Consider assigning roles more explicitly (solution preparer, timekeeper, observer, recorder) with role cards in future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	As the concentration of sodium thiosulfate solution decreased (more water was added), the time for the paper cross to disappear increased. The most concentrated solution (C1, full strength) produced the fastest reaction, while the most dilute solution (C4, 25 percent) produced the slowest reaction. When rate was calculated as 1 divided by time and plotted against concentration, the graph showed an approximately directly proportional increase — doubling the concentration roughly doubled the rate.
What did I learn?	Higher concentration means more solute particles are present in the same volume of solution. When there are more sodium thiosulfate particles per unit volume, they encounter HCl particles more frequently. More frequent collisions mean more effective collisions per second, so the sulfur precipitate forms faster and the cross disappears sooner. This is the same collision theory used in Lesson 3, but the variable is particle NUMBER rather than particle SPEED.
How does this explain the phenomenon?	The concentration effect explains why a more concentrated battery acid corrodes the metal terminals of a matatu faster than a dilute one — more acid particles per unit volume means more frequent collisions with the metal surface, producing corrosion products faster. It also explains why an antacid tablet dissolved in a small amount of water (more concentrated solution) may act faster than one dissolved in a full glass — and why pharmacists must carefully control the concentration of medicines to ensure the correct reaction rate inside the body.

LESSON 5 (80 minutes): Lesson 5: Surface Area and Reaction Rate — Grinding, Crushing, and the Secret Inside Solid Reactants

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate how surface area of a solid reactant affects the rate of a chemical reaction, and to connect particle-level reasoning to real-life practices in Kenyan homes, farms, and industries.
Knowledge	Learners explain that increasing the surface area of a solid reactant increases the rate of reaction by exposing more particles to collisions with reactant particles in solution or gas phase, thereby increasing collision frequency.
Skills	Learners plan and perform a controlled experiment comparing the reaction rate of marble chips (large pieces) versus marble powder with dilute hydrochloric acid, measuring gas volume or mass loss over time; they record, plot, and interpret results; they construct a particle-level diagram explaining their findings.
Attitudes	Learners appreciate the importance of manipulating surface area in biological, chemical, and physical processes in day-to-day Kenyan life (grinding maize at the posho mill, crushing limestone for cement at Bamburi, chewing githeri before swallowing, antacid tablets versus powder).
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	In Lessons 3 and 4 learners established that temperature and concentration speed up reactions by increasing collision frequency and energy. Lesson 5 introduces a third factor — surface area — while maintaining the same particle-collision logic. Learners now have three converging lines of evidence to update their class model and their Driving Question Board entry on what is happening at the particle level inside the thiosulfate beakers. They also begin to see that the same principle unifies a wide range of Kenyan contexts: the posho mill, the charcoal jiko, enzyme digestion of ugali, and limestone crushing at Bamburi Cement.
Safety Notes	Dilute HCl (1 mol/L or less): learners wear safety goggles and lab coats or aprons; no splashing; acid waste neutralised with sodium bicarbonate solution before disposal. Marble chips and powder: dust masks recommended when handling powder to avoid inhalation. No mouth contact with any reagents. Broken glassware disposed of in a designated sharps bin. Teacher demonstrates safe transfer of powder before learners proceed.

B. LESSON OVERVIEW

Learners investigate how surface area affects reaction rate using marble chips ($CaCO_3$) and dilute hydrochloric acid (HCl) — a classic, low-hazard system accessible in Kenyan school laboratories. The lesson opens with a Predict phase in which learners apply their existing particle-collision model (built in Lessons 3 and 4) to generate a new prediction. In the Observe phase, groups run parallel reactions using large marble chips, small marble chips, and marble powder against the same concentration of HCl at the same temperature, measuring either mass loss (using a balance) or CO_2 volume collected (using an inverted measuring cylinder over water). Data is tabulated and graphed. In the Explain phase, learners construct particle-level diagrams and connect findings to the anchoring phenomenon (why Beaker B's thiosulfate cross disappeared faster) and to Kenyan contexts. The DQB is updated with a new evidence card. In the Model Building phase, learners revise their cumulative particle-collision diagram to incorporate surface area alongside temperature and concentration. The lesson is designed for 80 minutes, accommodating practical setup and teardown.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Simple probability: yellow marble] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are presented with two sealed transparent bags placed at the front of the room: one contains large whole pieces of dried ugali (hardened maize cake), the other contains the same mass of finely ground ugali flour. The teacher asks: 'If I drop equal masses of these into a beaker of stomach acid, which one would dissolve faster?' Learners write a silent individual prediction in their science notebooks (3 minutes), using the sentence stem: 'I predict __ will react faster because, at the particle level, __.' Pairs then share predictions (Think-Pair-Share, 2 minutes). Selected pairs cold-called (4 pairs) share with the class. The teacher records all unique prediction reasons on the board without affirming or correcting. The teacher then pivots: 'Let us now think about our thiosulfate beakers. Both Beaker A and Beaker B used the SAME concentration and the sodium thiosulfate was already dissolved — so surface area of a solid was NOT the factor there. But today's question is: what happens when one of our reactants IS a solid? Does its size matter?' Learners add a new prediction row to their DQB prediction column.	1. Hold up the two bags of ugali simultaneously; say: 'Same mass. Same material. One difference — look carefully.' [WAIT 10 seconds for silent observation.] 2. Issue the prediction sentence stem on the board: 'I predict __ will react faster because, at the particle level, __.' Give 3 minutes of silent writing — circulate and note 3–4 interesting predictions to cold-call. 3. Say: 'Turn to your partner. You have 90 seconds. Share your prediction and your particle-level reason.' [WAIT 90 seconds.] 4. Cold-call exactly 4 pairs using name cards. After each response ask: 'Does anyone have a DIFFERENT particle-level reason?' Record all unique reasons on the board under the heading 'Our Predictions.' Do NOT evaluate. 5. Transition: 'Today we are testing surface area as our third factor. Let us connect this to our Driving Question — write a new prediction card for the DQB now.' [Give 2 minutes.]	Think-Pair-Share with Sentence Stems — the sentence stem forces learners to articulate a particle-level mechanism, not just a surface observation, activating prior knowledge from Lessons 3 and 4 (collision theory) and making implicit reasoning visible for the teacher's formative check.	Teacher circulates during silent prediction writing and notes whether learners reference 'more particles exposed,' 'more collisions,' or 'bigger surface' — these are the target ideas. Learners who write only 'it will be faster' without a particle-level reason are flagged for targeted questioning during the Explain phase. Observable indicator: at least 70% of learners' written predictions include a particle-level clause before the experiment begins.
Observe Phase 📹 **VIDEO:** [Simple probability: yellow marble] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups of 4 receive the following materials: dilute HCl (1 mol/L, 50 mL per trial), three pre-weighed samples of calcium carbonate — (A) 5 g large marble chips (~1–2 cm), (B) 5 g small marble chips (~3–5 mm), (C) 5 g marble powder — a 250 mL conical flask on a balance, a cotton wool plug, a stopwatch, and a data table in their	1. Before groups start, demonstrate the safe transfer of marble powder (hold flask low, pour slowly, cap immediately with cotton wool). Say: 'The cotton wool lets CO_2 escape but traps any acid mist. Your goggles stay ON for the entire practical.' 2. Start all groups simultaneously with the command: 'Pour your acid NOW — start your	Data Collection and Graphical Representation — learners generate their own quantitative evidence and immediately translate it into a visual rate-comparison graph. Drawing all three curves on one set of axes makes the effect of surface area directly perceptible, supporting pattern recognition before formal	Teacher checks that each group's data table has: correct units (grams, seconds), three columns (A, B, C), consistent reading intervals, and that mass loss is monotonically decreasing. Observable indicator: every group produces a completed data table with at least 8 time-points per trial. Teacher also spot-checks 3–4

📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	notebooks. Each group runs all three trials sequentially (or simultaneously if three setups are available). They record mass every 30 seconds for 5 minutes. They observe and note qualitative differences (speed of fizzing, sound, visible CO_2 bubbles). Groups also make a sketch of what they see in each beaker at the 1-minute mark. Data is recorded in a shared class table on the board. After all groups record, the teacher guides the whole class to draw a rate graph (mass lost on y-axis, time on x-axis, three curves on one set of axes, labelled A, B, C). Learners who finish early are directed to a challenge card: 'Predict the curve for marble dust so fine it is invisible — draw it on your graph and justify.'	stopwatch.' 3. Circulate every 3 minutes; at each group ask ONE targeted question rotating through: (a) 'What is the gas being produced? Write its formula.' (b) 'Which trial is losing mass fastest RIGHT NOW — how can you tell from the balance?' (c) 'At the particle level, WHY is the powder losing mass faster?' [WAIT 10 seconds after each question before accepting a response.] 4. At 18 minutes, call 'STOP — record your final mass and clean up safely. Neutralise acid with bicarbonate solution before pouring away.' 5. Compile the class data table on the board (2 minutes). 6. Instruct: 'Draw your three curves on one set of axes. Label each. You have 5 minutes.' Circulate and check axis labels, units, and curve shapes — correct errors immediately by asking: 'What does a steeper slope mean in terms of rate?'	explanation. The challenge card extension deepens reasoning for fast finishers by extrapolating the trend.	graphs for correct axis labelling and three distinguishable curves — groups with flat or identical curves are asked: 'What does it mean if two curves look the same? Does that match what you SAW in the flask?'
Explain Phase 📹 **VIDEO:** [Simple probability: yellow marble] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher presents three large diagrams on the board (or projected): a single large marble cube, the same cube cut into 8 smaller cubes, and the same mass ground into powder — each with the total surface area labelled (approximately 6 cm², 12 cm², and >100 cm² respectively, using simplified numbers). Learners are asked: 'Using your collision theory from Lesson 3, explain in writing why more surface area means more collisions.' They write a 3-sentence explanation in their notebooks using the structure: (1) Surface area statement, (2) Collision frequency statement, (3) Rate conclusion. Pairs share. The teacher cold-calls 3 learners to read their explanations aloud, then	1. Reveal the three surface area diagrams one at a time. After each, ask: 'How many faces are exposed to the acid? Count them.' [WAIT 10 seconds.] 2. Say: 'Write your three-sentence explanation NOW. You have 4 minutes. Silence.' Circulate — if a learner has not included a collision-frequency sentence, tap the board next to 'collision frequency' and ask quietly: 'Can you add this idea?' 3. Cold-call 3 learners to read aloud. After each, say: 'Class — thumbs up if this explanation includes a particle-level reason for MORE collisions. Thumbs sideways if it is partly there. Thumbs down if it is missing.' Count thumbs and report: 'I see 24 thumbs up — let us hear why two	Structured Three-Sentence Explanation with Peer Evaluation (Critique and Revision) — the constrained writing format ensures all three logical steps of the collision theory argument are present. Peer thumb-voting makes evaluation criteria explicit and gives the teacher an instant whole-class formative signal. The Kenyan context connections use analogical reasoning to transfer the abstract particle model to concrete, culturally familiar situations, deepening durable understanding.	Teacher reads 5–6 written explanations during silent writing and identifies the most common missing element (usually the collision-frequency link). This informs the cold-call selection — choose at least one explanation that IS missing the collision step so the class can improve it together. Observable indicator: after the class discussion, at least 80% of learners' written explanations contain all three required sentences. Teacher also listens for learners correctly distinguishing that surface area does NOT explain the thiosulfate phenomenon — this is a critical conceptual boundary.

	the class votes on which explanation is most complete using thumbs up/down/sideways. The class then connects the finding to four Kenyan contexts presented as quick image prompts on cards: (1) A posho mill grinding maize in Kisumu market — 'Why grind maize into flour before making ugali?' (2) A charcoal jiko using small charcoal pieces — 'Why does the jiko light faster with smaller charcoal pieces?' (3) A Bamburi Cement worker crushing limestone — 'Why does limestone need to be crushed before reacting with other materials?' (4) A person chewing githeri thoroughly — 'Why does your body need you to chew food before swallowing?' Groups discuss one context each (1.5 minutes) and report back in 30 seconds each. Finally, learners are asked: 'Go back to our thiosulfate phenomenon. The thiosulfate and HCl were BOTH dissolved. Does surface area explain the difference between Beaker A and Beaker B?' [Cold-call 2 learners.] Expected answer: No — surface area of dissolved particles is irrelevant; temperature was the key factor there. This consolidates the distinction between factors.	of you chose sideways.' 4. Distribute the four Kenya-context image cards to groups. Say: 'You have 90 seconds. Connect your surface area explanation to this context. One spokesperson — go.' Time strictly. After each group reports, add one sentence: 'At the particle level, more exposed surface means more collisions per second — so the rate is higher.' Repeat this phrase four times, once per context. 5. Ask the phenomenon reconnection question. [WAIT 15 seconds.] Cold-call 2 learners. Affirm the distinction clearly: 'Dissolved particles already have maximum contact — surface area is only a factor for SOLIDS. Well done for making that connection back to Beaker A and B.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Simple probability: yellow marble] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners individually complete a new DQB evidence card (sticky note or index card) using the three-column format established in Lesson 1: (1) WHAT WE OBSERVED today, (2) WHAT IT MEANS for our driving question, (3) NEW QUESTIONS this raises. They post their cards on the class DQB under the column 'Surface Area Evidence.' The teacher reads 3–4 cards aloud and facilitates a 2-minute whole-class discussion:	1. Distribute DQB cards. Say: 'Three columns — OBSERVED, MEANS FOR DRIVING QUESTION, NEW QUESTIONS. You have 3 minutes. Write in full sentences.' [WAIT 3 minutes.] 2. Collect or have learners post cards. Read 3–4 aloud — choose one that connects to the driving question well, one that raises a new question about catalysts or light, and one that contains a misconception (e.g., 'surface area	Driving Question Board Evidence Carding — the three-column structure (Observed → Meaning → New Questions) models scientific reasoning by requiring learners to connect empirical evidence to explanatory claims and to identify the boundaries of their current understanding. Posting cards publicly makes the class's growing knowledge base visible and motivates curiosity about upcoming lessons (catalysts, light).	Teacher reads all posted cards before Lesson 6 to identify: (a) learners who connect surface area to collision frequency (target understanding), (b) learners who conflate surface area with temperature or energy (common misconception to address in Lesson 6 warm-up), (c) new questions that can be used as hooks for Lessons 6 and 7. Observable indicator: every learner posts a card with at least

📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'Look at our board. We now have evidence from temperature (Lesson 3), concentration (Lesson 4), and surface area (today). What pattern do you notice across all three factors?' Learners articulate the unifying idea: all three factors increase the rate by increasing collision frequency and/or collision energy. The teacher writes on the board: 'Our Growing Claim: Reaction rate increases when collision frequency or collision energy increases.' Learners copy this into their notebooks as the current class consensus model statement. New questions generated by learners (e.g., 'What if we used a catalyst — does that also increase collisions?') are noted on the DQB as 'Questions for Future Lessons.'	gives particles more energy') to address gently. 3. Ask the synthesis question: 'Three lessons, three factors — what is the ONE thing they all do?' [WAIT 15 seconds. Cold-call 3 learners.] 4. Write the Growing Claim on the board. Say: 'Copy this. This is our best scientific explanation RIGHT NOW — it may grow in the next two lessons.'		two of the three columns completed in writing.
Model Building Phase 📹 **VIDEO:** Simple probability: yellow marble Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative particle-collision diagrams begun in Lesson 3 and revised in Lesson 4. Today they add a third panel to their diagram: 'Factor 3 — Surface Area.' The panel must show (a) a large marble chip with HCl particles colliding only at its outer surface, and (b) the same mass of marble powder with HCl particles colliding at many more exposed surfaces simultaneously. Arrows indicate CO_2 molecules being produced. Learners annotate: 'More surface area → more particle collisions per second → faster rate.' Pairs compare diagrams and check each other's using the class 'Model Quality Checklist' (posted on the wall): ✓ Particles shown as circles, ✓ Collisions shown at the surface only (not inside the solid), ✓ Products labelled (CO_2, water, $CaCl_2$), ✓ Comparative annotation	1. Say: 'Take out your cumulative model from Lesson 3. You are adding Panel 3 today — Surface Area. You have 8 minutes.' [Display the Model Quality Checklist on the board.] 2. Circulate and look specifically for learners drawing HCl particles colliding with the INTERIOR of the marble chip — this is a common error. If observed, say: 'In a solid, can HCl particles get INSIDE? Think about what a solid lattice looks like.' [WAIT 10 seconds.] 3. At 7 minutes, instruct pairs: 'Swap diagrams. Check against the checklist. Put a tick next to each criterion your partner has met. If something is missing, circle it gently.' [1 minute.] 4. At 8 minutes, ask 2–3 pairs to hold up their diagrams for a gallery walk (learners stand and look at 2 nearby diagrams in 30 seconds). 5. Close: 'In Lesson 6 we will add	Cumulative Model Revision — by physically extending the same diagram across lessons, learners build a coherent, growing explanatory model rather than isolated lesson-by-lesson notes. Peer checklist review introduces scientific critique norms (evaluating a model against criteria) and helps learners identify gaps in their own representations before the teacher does.	Teacher collects or photographs 6–8 models during the gallery walk and reviews them before Lesson 6. Look for: (a) collisions correctly drawn at surface only, (b) correct products labelled (CO_2 bubbles rising), (c) comparative annotation explicitly linking more surface to more collisions. Models that show collisions inside the solid are flagged for a targeted warm-up correction at the start of Lesson 6. Observable indicator: at least 85% of diagrams correctly depict surface-only collisions with a comparative annotation present.

	present. Groups who finish early are asked to add a 'Real-Life Connection' label linking one of today's Kenyan contexts to their diagram (e.g., draw a small charcoal piece and label it 'Jiko — small charcoal = more surface area = faster combustion').	Factor 4 — catalysts. Think about this tonight: if a catalyst speeds up a reaction WITHOUT being used up, what might it be doing at the particle level? Write one hypothesis in your notebook before next class.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. CONCEPTUAL BOUNDARY: Did learners successfully distinguish that surface area applies to SOLID reactants and does NOT explain the thiosulfate phenomenon (where both reactants were dissolved)? If learners conflated this, plan a 3-minute warm-up correction for Lesson 6 using a direct comparison diagram.
2. PRACTICAL MANAGEMENT: Was the mass-loss method reliable with available balances? If balances were insufficiently sensitive (±0.1 g or worse), consider switching to the inverted cylinder gas-collection method for future iterations, or pre-testing to confirm measurable mass loss within 5 minutes using the available equipment.
3. KENYAN CONTEXT UPTAKE: Which of the four context cards (posho mill, jiko, Bamburi Cement, chewing githeri) generated the richest discussion? Use the strongest context as a recap hook at the start of Lesson 6.
4. MODEL QUALITY: Review the 6–8 photographed diagrams. Are learners consistently drawing collisions at the surface rather than the interior? If the interior-collision error is widespread (>30% of models), design a brief 'What is a solid lattice?' micro-explanation for Lesson 6.
5. DQB PROGRESSION: Are learners beginning to articulate the unifying claim (collision frequency/energy) across all three factors independently, or are they treating each factor as isolated? If isolated thinking persists, consider opening Lesson 6 with a class discussion explicitly comparing the three evidence cards side by side before introducing the catalyst factor.
6. PACING: The practical phase is the most time-sensitive. If groups were not finished by the 18-minute mark, identify whether the cause was setup time, data recording, or group management — and adjust the pre-lab briefing time accordingly for Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Marble powder reacted with dilute HCl faster than small chips, which reacted faster than large chips — all at the same temperature and concentration. The powder flask showed rapid fizzing and the fastest mass loss. The large-chip flask showed slow, steady bubbling. All three eventually produced the same total mass loss (same amount of reactant used up), but the RATE differed dramatically.
What did I learn?	Increasing the surface area of a solid reactant increases the rate of reaction because more particles of the solid are exposed to collisions with the acid particles at any given moment, increasing collision frequency. This is the same collision-frequency principle seen with temperature (Lesson 3) and concentration (Lesson 4). Surface area is only relevant for SOLID reactants — dissolved reactants already have maximum particle exposure.
How does this explain the phenomenon?	At the particle level, a large marble chip exposes only its outer layer of $CaCO_3$ particles to HCl(aq) particles. When the same mass is ground to powder, millions of new surfaces are created, exposing vastly more $CaCO_3$ particles simultaneously. More exposed particles means more successful collisions per second, so CO_2 and $CaCl_2$ are produced at a higher rate. This explains why grinding maize at the posho mill, crushing charcoal for the jiko, breaking limestone at Bamburi Cement, and chewing githeri

	thoroughly all use the same principle to speed up chemical or biological reactions in everyday Kenyan life.

LESSON 6 (40 minutes): Factor 4: Catalysts and Light — Speeding Up Reactions Without Being Used Up	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will understand how catalysts and light act as additional factors that control reaction rate, and connect these factors to Kenyan industrial and biological processes.
Knowledge	Students will be able to: (1) define a catalyst as a substance that increases the rate of a reaction without being consumed; (2) explain how enzymes function as biological catalysts using the liver-and-hydrogen-peroxide demonstration; (3) distinguish between catalysts and reactants; (4) describe the role of the iron catalyst in the Haber process for ammonia and fertiliser production at Athi River; (5) explain how light can increase the rate of certain reactions, using photosynthesis and photographic film as examples.
Skills	Students will practise: careful observation and recording of a teacher demonstration; comparing experimental outcomes (fresh liver vs boiled liver vs sand); constructing written and diagrammatic explanations using collision theory and activation-energy concepts; researching and presenting findings about one Kenyan industrial or biological process that controls reaction rate.
Attitudes	Students will appreciate the value of enzymes and catalysts in sustaining life and economic productivity in Kenya; they will demonstrate intellectual curiosity by connecting laboratory observations to real-world applications; they will show responsibility by citing sources accurately during their research activity.
Key Inquiry Question	How do various factors affect the rate of reactions? — specifically: how do catalysts and light alter reaction rate without changing reactant identity or concentration?
Purpose in Storyline	Lessons 3, 4, and 5 established that temperature, concentration, and surface area all increase reaction rate by increasing the frequency or energy of particle collisions. Lesson 6 adds catalysts and light as two more control levers — ones that work differently: a catalyst lowers the energy barrier for collisions to be successful, while light supplies energy directly to reactant particles. These two factors complete the set of rate-controlling variables students need in order to build a full collision-theory model in Lesson 7 and the synthesis in Lesson 8. The liver demonstration also reactivates the biological dimension of the anchoring phenomenon: just as warm Mombasa temperatures sped up the thiosulfate reaction by energising particles, enzymes in living cells speed up reactions by a different mechanism — reducing the energy threshold — which explains digestion, fermentation of uji, and metabolic processes in the gut.
Safety Notes	Hydrogen peroxide (H2O2) at 3% (pharmacy-grade) is used — teacher demonstration only, not student handling. Avoid skin and eye contact; teacher wears gloves and safety goggles. Liver pieces should be handled with tongs; wash hands after any contact. Boiled liver preparation done before the lesson. No open flames during the demonstration. Broken glassware should be disposed of in a designated sharps container. If manganese(IV) oxide (MnO2) is used as an alternative catalyst, handle as fine powder carefully to avoid inhalation.

B. LESSON OVERVIEW

In this 40-minute lesson the teacher opens by connecting back to the anchoring phenomenon: we know temperature, concentration, and surface area all change how often particles collide — but are there ways to speed up reactions that do NOT depend on those three factors? Students predict what will happen when pieces of fresh liver are added to hydrogen peroxide, then observe the dramatic fizzing (oxygen gas released) alongside the flat results for boiled liver and inert sand. They explain the difference using the concept of the enzyme catalase — a biological catalyst that lowers the energy needed for successful collisions without itself being consumed. The class then extends this to industrial catalysts (iron in the Haber process, relevant to fertiliser production for Kenyan farmers) and to light-driven reactions (photosynthesis in Kenyan tea and maize crops; historical photographic film). A structured 8-minute research-and-share activity lets groups investigate optimum conditions in one Kenyan process. The lesson closes with a DQB update: students add new evidence cards and revise their particle-collision models to include the activation-energy idea.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students open their science notebooks and read back the summary sentence they wrote at the end of Lesson 5: increasing surface area exposes more solid particles to collisions, increasing collision frequency. The teacher then poses the challenge question: suppose you want to speed up a reaction but you cannot change the temperature, the concentration, OR the surface area — is there any other way? Students write a 2-minute individual prediction in their notebooks, drawing on prior knowledge of digestion, cooking, or any industrial process they know. Volunteers share predictions; the teacher records key ideas on the board under the heading What ELSE could change reaction speed? without confirming or correcting. Students then observe three test tubes prepared on the demonstration bench: Tube 1 contains fresh raw liver (about 2 cm3); Tube 2 contains boiled liver (same size, same mass, boiled for 5 minutes); Tube 3 contains a similar volume of clean sand. Students predict — by writing in notebooks — which tube will react most vigorously when hydrogen peroxide is added, and why. They sketch the three tubes and annotate their expected outcome.	Teacher displays the three prepared tubes and asks: Before I add anything, look carefully at what is in each tube. What is the same? What is different? WAIT 20 SECONDS. After students respond, teacher says: I am going to add the same volume of the same hydrogen peroxide solution to each tube at the same time. Your job is to watch closely — note what you see, hear, and smell. Record EVERYTHING. Then teacher asks: On a scale of 1 to 5 — 1 meaning no reaction, 5 meaning very vigorous — write your prediction for each tube right now. Seal it. Do not change it after you observe. WAIT 30 SECONDS for silent individual writing. Teacher checks that all students have written a prediction before proceeding.	Prediction journaling activates prior knowledge of biological systems (digestion, fermentation) and creates cognitive dissonance when the boiled-liver result confounds expectations of those who predicted it would behave like the fresh liver. The three-tube comparison isolates the variable (enzyme activity) and prefigures the controlled-experiment logic students have been practising since Lesson 2.	Teacher circulates and reads predictions over shoulders. Look for students who already mention enzymes or proteins — these can be highlighted later as bridge-builders. Note students who predict sand will react — a useful misconception to address in the Explain phase. No marks are given; the purpose is diagnostic.
Observe Phase 📹 **VIDEO:** Common	The teacher adds approximately 5 mL of 3% hydrogen peroxide simultaneously to each of the three tubes. Students observe for 90	After adding hydrogen peroxide, teacher steps back and says: Observe silently for 30 seconds — no talking. Just watch. WAIT 30	Structured observation table scaffolds noticing at multiple time points, preventing students from writing only a single-word	Teacher checks whether students correctly identified Tube 1 as most reactive and Tube 2 as least reactive (among the liver tubes).

polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	seconds and record in a structured table in their notebooks: (Column 1) Tube contents; (Column 2) Observations during first 30 seconds; (Column 3) Observations at 60 seconds; (Column 4) Any gas produced? (test with glowing splint if time allows); (Column 5) How does this compare to your prediction? The expected results are: Tube 1 (fresh liver) — vigorous bubbling, froth rises, oxygen gas detected by re-lighting a glowing splint; Tube 2 (boiled liver) — very little or no bubbling; Tube 3 (sand) — no reaction. Students circle the tube that surprised them most and write one sentence explaining why they were surprised. Class data is called out and recorded on the shared class whiteboard so all groups can see the pattern across the three tubes.	SECONDS. Then: Now record what you see. You have 60 seconds. WAIT 60 SECONDS. Teacher then holds a glowing splint above Tube 1 to test for oxygen, re-igniting it, and narrates: The splint re-lights — that tells us a gas that supports combustion is being produced. What gas could that be? WAIT 15 SECONDS. Teacher then asks: Tube 2 has exactly the same liver proteins as Tube 1 — same animal, same organ, same mass. Why did Tube 2 behave so differently? I want you to think about this before you answer. WAIT 25 SECONDS. Teacher collects two or three student responses and writes them verbatim on the board without editing.	observation. The glowing splint confirms a testable claim (oxygen produced), modelling scientific evidence standards. Writing a surprise sentence forces metacognition: comparing prediction to outcome is the core of evidence-based reasoning.	Students who predicted Tube 2 would be equally reactive as Tube 1 reveal a misconception that heating does not affect protein structure — this is addressed explicitly in the Explain phase. The glowing-splint test gives a concrete, memorable anchor for the claim that oxygen was produced.
Explain Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students attempt a written explanation in their notebooks using the following sentence scaffold provided on the board: Fresh liver contains a biological catalyst called _______, which _______ the activation energy needed for hydrogen peroxide to decompose into _______ and _______. Boiled liver cannot catalyse the reaction because _______. A catalyst is different from a reactant because _______. After 3 minutes of individual writing, pairs share and refine their explanations. The teacher then delivers a 6-minute structured explanation covering four ideas: (1) Catalase is an enzyme — a protein molecule — found in liver cells (also in potato, papaya, and the leaves of Kenyan tea plants); it provides an alternative reaction pathway that requires less energy	Before revealing the word enzyme, teacher asks: Has anyone heard the word enzyme before? Where? WAIT 15 SECONDS. After collecting responses — likely from biology lessons, digestion, washing powder adverts — teacher says: Good. Today we are going to see the same idea from a chemistry angle — enzymes are catalysts, and catalysts are a chemistry concept. They are not magic. They work because of particle-level chemistry. Let us build the explanation together. During the explanation of denaturation, teacher pauses and asks: If boiling destroyed the enzyme, what does that tell us about how temperature affects biological catalysts compared to how it affects simple chemical reactions? Think for 20 seconds before you answer. WAIT 20	The sentence scaffold reduces cognitive load while maintaining productive struggle: students must supply the key terms rather than copy a complete sentence. The industrial parallel (Haber process, Athi River) grounds an abstract chemistry concept in Kenyan economic reality. The light-reaction discussion broadens the factor list and invites students to generate Kenyan examples, reinforcing transfer of learning. The OPTIMUM temperature prompt intentionally creates a bridge to Lesson 7, maintaining storyline coherence.	Teacher reads completed sentence scaffolds from at least four students (two who were on-track in the Predict phase, two who showed misconceptions). Key checkpoints: Can the student name catalase? Can the student state that the catalyst is unchanged after the reaction? Can the student articulate WHY boiling destroyed the enzyme? Can the student give one Kenyan example of a light-sensitive reaction? Students who cannot distinguish catalyst from reactant are noted for targeted support in Lesson 7.

	to get started. (2) Boiling denatures the enzyme — the protein chain unfolds and loses its shape, so it can no longer work. This is why the boiled liver did nothing. (3) A catalyst is NOT consumed in the reaction: after it has helped one set of hydrogen peroxide molecules decompose, catalase is free to help the next set. This is fundamentally different from a reactant, which is used up. (4) Industrial parallel: in the Haber process, iron (or iron oxide) acts as a catalyst for the reaction between nitrogen gas and hydrogen gas to make ammonia — the raw material for fertilisers used by Kenyan wheat and tea farmers. The plant at Athi River (East African Portland Cement area, Machakos County) is an example of industrial chemistry that depends on catalysts to make fertiliser affordable. Students add a labelled diagram to their notebooks showing: reactants entering a catalyst surface, products leaving, and the catalyst surface unchanged. The class then briefly discusses light as a rate-controlling factor: light provides energy that can initiate or accelerate certain reactions. Examples given: (a) photosynthesis in Kenyan maize and tea — light energy drives the reaction in chloroplasts; (b) old photographic film — silver halide salts decompose when exposed to light (students may have seen old family photos fade); (c) some medicines must be stored in dark amber bottles because light degrades them. Teacher asks: Can you think of any Kenyan situation where keeping a chemical AWAY from light would	SECONDS. This question is designed to surface the idea that enzymes have an OPTIMUM temperature — above which they are destroyed — unlike inorganic reactions where higher temperature always means faster reaction. Teacher then says: This distinction will be critical in Lesson 7 when we look at digestion and fermentation. Write the word OPTIMUM in your notebook and put a star next to it.		

	slow down an unwanted reaction? WAIT 20 SECONDS. Expected responses: storing cooking oil away from sunlight to prevent rancidity; storing medicines; keeping germination experiments in the dark.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the Driving Question Board displayed at the front of the room. The teacher hands each student two sticky notes of different colours: one colour for NEW EVIDENCE (what we now know) and one colour for NEW QUESTIONS (what we still wonder about). Students write independently for 2 minutes. Likely NEW EVIDENCE cards: catalysts speed up reactions without being used up; enzymes are biological catalysts; boiling destroys enzymes; iron is the catalyst in the Haber process; light speeds up some reactions like photosynthesis. Likely NEW QUESTIONS cards: what is activation energy exactly? Do all reactions have catalysts? Can a catalyst slow down a reaction? Why does the human body need enzymes if reactions happen anyway? What happens to catalysts in the body after they work? What other industrial processes use catalysts in Kenya? Students post their cards on the DQB. The teacher reads three or four cards aloud and clusters them into the emerging categories: Biological catalysts (enzymes); Industrial catalysts; Light-driven reactions; Still-unexplained questions. Teacher then connects back to the anchoring phenomenon: In our cold-vs-warm thiosulfate beakers at the start of this unit, we explained the	Teacher reads one particularly interesting NEW QUESTIONS card aloud and says: This is a great question. It is not one we can answer today, but it is exactly the kind of question a KEMRI biochemist or an industrial chemist at Athi River would ask every day. Who wrote this? (Pause for hand raise.) Can you say more about what you are wondering? WAIT 15 SECONDS. Teacher affirms the question and leaves it visibly posted on the DQB, NOT answered. This models that science produces new questions as well as answers.	The two-colour sticky-note system visually separates evidence from inquiry, reinforcing the distinction between what has been established and what remains open. Clustering cards in real time shows students how scientific questions organise into themes. Re-connecting to the thiosulfate phenomenon keeps the anchoring event alive and prevents knowledge from feeling disconnected from its origin.	Teacher scans evidence cards for accuracy: any card that misrepresents a catalyst (for example, a card that says catalyst is used up) is gently challenged in the moment — teacher picks it up and asks the class: Does everyone agree with this? What do we know about whether a catalyst is consumed? WAIT 15 SECONDS. This is not punitive; it is a class-level formative correction.

	difference using TEMPERATURE. But now imagine: what if instead of changing the temperature of Beaker B, we had added a tiny amount of a catalyst? Would the same cross disappear faster? Students give a show-of-hands vote: yes / no / unsure. Teacher records the vote and says: Hold that thought. We will use it in Lesson 7.			
Model Building Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 8 minutes, students open to a dedicated Model page in their notebooks (begun in Lesson 1 and updated each lesson). They add a new panel labelled Lesson 6: Catalysts and Light. In this panel they draw: (a) a reaction energy diagram (simplified — a hill showing activation energy with and without a catalyst, labelled Reactants at start, Products at end, Activation energy barrier, and Catalyst lowers the barrier); (b) a brief annotation connecting this diagram to both the liver demonstration and the Haber process; (c) a one-sentence statement about light: Light provides energy that can lower the barrier for light-sensitive reactions, increasing the chance that collisions lead to successful reactions — examples: photosynthesis in Kenyan tea, decomposition of medicines in sunlight. Students then update the cumulative rate-factors table they have been building since Lesson 3, adding a new row: Factor — Catalyst (or Light); What changes at the particle level — lowers activation energy so more collisions are successful even at the same temperature and concentration; Kenyan example — catalase in uji fermentation, iron	Teacher circulates during model building and prompts with two questions: (1) Your model should show that without the catalyst, most collisions fail even if they happen. Can you show that idea on your diagram? WAIT 20 SECONDS per pair. (2) Look at the row in your table for temperature (Lesson 3) and the row for catalysts (today). What is the key difference in HOW they each speed up the reaction? WAIT 20 SECONDS. At the 2-minute warning, teacher asks the whole class to stand and share their Kenyan example from the table row with the person next to them — 30 seconds only. Teacher then closes the lesson with: You now have four factors in your collision-theory toolkit: temperature, concentration, surface area, and catalysts or light. In Lesson 7 we are going to use ALL FOUR to explain some of the most important processes in Kenyan life — from the fermentation jar of uji on a Kisumu kitchen shelf to the rusting iron-sheet roofs in Mombasa. Start thinking about which factor matters most in each of those situations.	The cumulative model page makes learning visible across lessons and supports longitudinal sense-making. The activation-energy diagram introduces a new representational form (energy diagram) that deepens conceptual understanding beyond the particle-collision picture. The teacher-prompted comparison between temperature and catalysts consolidates the key distinction and prepares students for the synthesis lesson. The 30-second pair-share on Kenyan examples is a low-stakes retrieval practice that reinforces transfer.	Teacher looks for two key ideas in the model: (1) the catalyst lowers the activation energy hill (not increases collisions); (2) the Kenyan example is genuinely connected to the factor being described. Students who draw a catalyst increasing collision frequency (a common misconception) are identified and a correction is planted verbally — teacher says quietly to those pairs: Check your diagram again — does a catalyst add more particles to the mixture, or does it give the existing particles a lower hill to climb? That distinction matters. These students are flagged for follow-up at the start of Lesson 7.

	catalyst in fertiliser production, light in photosynthesis of maize. Pairs compare their updated models and check for consistency: does their model correctly show that a catalyst does NOT change the number of collisions (unlike temperature or concentration) but changes the fraction of collisions that are successful? This distinction is highlighted by the teacher with a brief whole-class pointer before the lesson ends.			

D. TEACHER REFLECTION

After the lesson, consider: (1) Did students successfully distinguish a catalyst from a reactant, or did the most common confusion persist? If so, plan a 2-minute re-teach at the start of Lesson 7 using the analogy of a mountain pass — a shortcut that reduces the height climbers must scale, but the pass itself is not consumed. (2) Were the Kenyan industrial examples (Haber process, Athi River, KEMRI) meaningful to students or did they feel abstract? If abstract, consider bringing a fertiliser packet (e.g., CAN or DAP) to Lesson 7 so students can read the nitrogen content and connect it physically to the Haber process. (3) Did the DQB generate genuine new questions, or did students write generic statements? Genuine new questions suggest high engagement; generic statements suggest the activity was treated as a task to complete rather than a thinking tool — adjust by modelling a specific, curious question yourself next time. (4) Did the cumulative model table remain a meaningful tool or has it become mechanical copying? Check three or four tables after the lesson for quality of explanation in the particle-level column — this is the most important column for assessing conceptual understanding. (5) Time check: if the lesson ran long on the demonstration, the research-and-share activity described in the storyline overview can be moved to the first 5 minutes of Lesson 7 as a warm-up presentation, ensuring no student group loses the opportunity to present their Kenyan process findings.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Fresh liver added to hydrogen peroxide produced vigorous bubbling and oxygen gas. Boiled liver produced no detectable reaction. Sand produced no reaction. The same hydrogen peroxide was used in all three tubes.
What did I learn?	A catalyst speeds up a reaction without being used up. Enzymes are biological catalysts that work by lowering activation energy. Heating destroys enzyme activity. Iron catalysts are used industrially in fertiliser production. Light energy can drive or accelerate certain reactions.
How does this explain the phenomenon?	Catalysts work by providing an alternative reaction pathway with a lower activation energy barrier, meaning a greater fraction of particle collisions have enough energy to be successful — even though the total number of collisions has not changed. This is different from temperature (which increases collision frequency and energy) and from concentration or surface area (which increase collision frequency). Enzymes achieve this through their precise three-dimensional shape, which is destroyed by high temperatures. Light-sensitive reactions are accelerated because light supplies energy directly to reactant particles, increasing the proportion of successful collisions. All these factors — temperature, concentration, surface area, catalysts, and light — ultimately determine whether the particles inside a reaction vessel collide with sufficient energy and correct orientation to form new products, explaining everything from the disappearing thiosulfate cross to the rising froth of uji on a Kisumu stove.

LESSON 7 (80 minutes): Light and Catalysts — The Final Rate Factors	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how light intensity and catalysts affect the rate of chemical reactions, and to connect these factors to the particle-collision model built across the unit.
Knowledge	Learners describe how catalysts lower activation energy to increase reaction rate without being consumed, and how light provides energy to initiate photochemical reactions.
Skills	Learners design and carry out controlled experiments to investigate the effect of a catalyst (manganese dioxide) on hydrogen peroxide decomposition and observe a light-sensitive reaction; they record, tabulate and interpret results using the collision theory framework.
Attitudes	Learners appreciate the importance of catalysts and light in biological, industrial and everyday Kenyan contexts — from enzyme action in the human gut to photosynthesis in maize fields and the manufacture of fertiliser at Kenya Pipeline depots.
Key Inquiry Question	How do various factors affect the rate of reactions?
Purpose in Storyline	Lessons 1–6 established that temperature, concentration, and surface area each speed up reactions by increasing the frequency or energy of collisions between particles. Lesson 7 closes the evidence-gathering sequence by adding the final two factors — catalysts and light — completing the set of tools learners need to fully explain the anchoring phenomenon and to answer the driving question about controlling reaction speed in Kenyan life.
Safety Notes	Hydrogen peroxide (20 vol) is an oxidising and skin/eye irritant — learners must wear safety goggles and gloves throughout. Manganese dioxide is non-toxic in small quantities but must not be ingested; dispose in the waste beaker provided. No open flames during the hydrogen peroxide experiment. If hydrogen peroxide contacts skin, rinse immediately with plenty of water for 15 minutes and report to the teacher. Keep the silver nitrate / photographic paper activity away from direct sunlight until the moment of investigation.

B. LESSON OVERVIEW
Lesson 7 of 8 in the evidence-gathering sequence. Learners investigate the two remaining rate factors — catalysts and light — through two focused practical stations. In Station A, groups test the effect of manganese dioxide (MnO_2) on the decomposition of hydrogen peroxide ($H_2O_2 \rightarrow H_2O + O_2$), using a glowing splint to confirm oxygen release and timing the vigour of effervescence. In Station B, groups expose light-sensitive photographic paper (or a silver nitrate–coated card) to varying light intensities (torch/sunlight/shade) and observe colour change rate. Both data sets are interpreted using the collision theory model, linking catalysts to lowered activation energy and light to photon-driven bond breaking. All findings are mapped back to the anchoring thiosulfate phenomenon and to Kenyan real-world contexts: enzyme catalysts in uji fermentation and githeri digestion, chlorophyll and photosynthesis in Rift Valley tea and maize crops, and industrial catalysts in fertiliser manufacture. By the end of the lesson, learners update their particle-level models and populate the penultimate row of the Driving Question Board, leaving only the synthesis lesson (Lesson 8) to integrate all factors.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners open their science	Distribute scenario cards as	Prediction Journaling with	Observable indicator: Every

📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Equivalent Rates (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	notebooks to a fresh page headed 'Lesson 7: Catalysts and Light.' They read two short scenario cards placed on their desks: – Scenario A: 'A factory in Eldoret uses manganese dioxide powder to speed up the breakdown of hydrogen peroxide when making oxygen for medical use. When MnO_2 is added, the liquid fizzes violently. After the reaction, the MnO_2 is recovered and used again the next day.' – Scenario B: 'Photographs left in bright Nairobi sunlight fade within hours. The same photographs stored in a dark drawer stay sharp for years.' Learners individually write predictions in their notebooks for two questions: 1. How does MnO_2 speed up the reaction — does it get used up? What happens to the particles? 2. Why does light cause chemical change in the photograph — and could light speed up or slow down other reactions? Learners then share predictions with a partner using a Think-Pair-Share protocol before one pair is cold-called per table.	learners settle (0–2 min). Read Scenario A aloud using a measured voice; pause and say: 'Notice the phrase — used again the next day. What does that tell you about manganese dioxide?' Allow 10 seconds of wait time. Do NOT answer. Move to Scenario B; read aloud and ask: 'We have already shown that temperature, concentration, and surface area all change reaction speed by changing collisions. How could LIGHT possibly do the same thing?' Allow 15 seconds of wait time. Circulate silently for 4 minutes while learners write individual predictions — resist the urge to hint. After pair-share (2 min), cold-call exactly 3 pairs: one confident pair, one hesitant pair, one pair with a surprising prediction. Record key prediction phrases verbatim on the chalkboard under two columns: 'Catalyst predictions' and 'Light predictions.' Do not evaluate correctness — say only: 'Interesting — let's gather evidence to test that.'	Anchored Scenarios — by grounding predictions in concrete Kenyan industrial and everyday contexts (Eldoret factory, Nairobi photographs), learners activate prior knowledge of collision theory from Lessons 3–6 and generate personal hypotheses they are motivated to test. Recording predictions publicly on the board creates cognitive commitment that deepens attention during the observation phase.	learner has written at least two sentences of prediction in their notebook before pair-share begins. Teacher scans notebooks during circulation — flag any learner with a blank page for immediate quiet prompting. During cold-call, listen for whether learners spontaneously use collision-theory language ('particles,' 'energy,' 'activation energy,' 'frequency of collisions'); note on a mental register which groups are ready to extend and which need scaffolding during the experiment.
Observe Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of	The class is divided into six groups of ~5. Each group rotates through two practical stations (20 min each, with a 2-min changeover and equipment wipe-down). STATION A — The Catalyst Station (MnO_2 and H_2O_2): Materials per group: two identical 100 mL conical flasks, 20 mL of 20-vol hydrogen peroxide in each, a spatula of manganese dioxide	Before stations open (2 min): Stand at the front and deliver the safety briefing verbatim: 'Hydrogen peroxide is an oxidising agent — it will irritate your skin and eyes. Goggles and gloves ON before you touch any reagent. If any liquid touches your skin, go to the sink immediately and rinse for 15 minutes. MnO_2 goes into the waste beaker — not the sink.' Physically demonstrate putting on goggles. Assign a Safety Officer in each	Dual-Station Structured Inquiry (NGSS SEP 3 & 4) — running two parallel investigations within the same lesson allows learners to gather evidence for two distinct mechanisms (catalytic lowering of activation energy; photon-driven bond breaking) and immediately begin cross-comparing them. The structured results table scaffolds data recording without removing the cognitive challenge of interpretation. Requiring learners	Observable indicator: Each learner's notebook contains a completed two-column results table for Station A and at least three time-point colour sketches for Station B before Phase 3 begins. Teacher collects a quick show-of-hands exit check at the changeover: 'Thumbs up if your glowing splint relighted in Flask 2.' If fewer than 60% of groups report a positive result, troubleshoot immediately (likely cause: H_2O_2

[Equivalent Rates (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	(MnO_2) powder, a stopwatch, two glowing splints, a ruler, safety goggles, gloves, a waste beaker. Procedure: 1. Flask 1 (Control): Observe 20 mL H_2O_2 alone for 60 seconds. Hold glowing splint above mouth — record observation. 2. Flask 2 (Catalyst): Add one spatula of MnO_2 to 20 mL H_2O_2. Start stopwatch. Observe and record: (a) time for visible vigorous effervescence to begin, (b) whether the glowing splint relights, (c) whether the MnO_2 appears to be consumed. 3. After the reaction slows, filter the MnO_2 onto filter paper — observe and record its appearance and mass (qualitative comparison). Learners record all observations in a two-column results table: 'Flask 1 — no catalyst' vs. 'Flask 2 — with MnO_2.' STATION B — The Light Station (Photosensitive paper): Materials per group: three strips of photosensitive paper (or silver nitrate–coated card) in a light-proof envelope, a bright torch, a watch, a sheet of black card to create shade. Procedure: 1. Strip 1 (Dark control): Keep inside envelope. Open at end of experiment to observe colour. 2. Strip 2 (Dim light): Hold 30 cm from torch for 60 seconds. Observe and record colour change at 20 s, 40 s, 60 s. 3. Strip 3 (Bright direct sunlight or torch at 5 cm): Expose for 60 seconds. Observe and record colour change at 20 s, 40 s, 60 s.	group whose job is to enforce PPE. During Station A rotations: Circulate continuously. At each group, ask exactly two probing questions — choose from: 'What is the gas being produced — how do you know?' (wait 10 s); 'Is the MnO_2 disappearing — what does that tell you?' (wait 12 s); 'If this were happening inside your body — which molecules in your cells do the same job as MnO_2?' (wait 15 s). Do NOT give answers. During Station B rotations: Ask: 'What is different about the energy the particles in Strip 3 are receiving compared to Strip 1?' (wait 12 s). Prompt learners to connect light to the collision model: 'Can a photon — a packet of light — give energy to a particle the way heat does?' (wait 15 s). At the 2-minute changeover bell: Say loudly: 'Freeze — record your final observation NOW before you move. Safety Officers, check PPE is removed before groups stand.'	to observe the recovered MnO_2 after the reaction is the key empirical hook for the concept of a catalyst not being consumed.	has degraded — replace stock). Listen for learners who spontaneously state 'the MnO_2 was still there after the reaction' — these learners are ready for the activation energy extension in Phase 3.

	Learners sketch the colour of each strip at each time point in their notebooks. All observations are recorded in notebooks with sketches and data tables as per NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 4 (Analysing and Interpreting Data).			
Explain Phase 📹 **VIDEO:** Common polyatomic ions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Equivalent Rates (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their home seats. The teacher projects (or draws) two labelled energy diagrams on the board: – Diagram 1: Reaction energy profile WITHOUT a catalyst — showing the high activation energy 'hill' that reactant particles must climb. – Diagram 2: The same reaction WITH a catalyst — showing a lower activation energy hill (alternative reaction pathway). Learners annotate both diagrams in their notebooks, labelling: reactants, products, activation energy (E_a), and the effect of the catalyst on E_a. The class then engages in a Socratic Seminar (10 min) on three connecting questions posed by the teacher: 1. 'In Station A, why did the reaction happen faster WITH MnO_2, even though we did not heat the flask — what did MnO_2 change about the energy landscape of the reaction?' 2. 'In Station B, what role did light play — and how is that similar to, or different from, the role of heat in the thiosulfate experiment we	Draw both energy diagrams slowly on the board, narrating: 'This hill represents the activation energy — the minimum energy particles need for a successful collision. Without a catalyst, the hill is HIGH — few particles have enough energy to get over it.' (Pause 10 s.) 'The catalyst provides an alternative pathway — a tunnel through the hill — so more particles can react at the SAME temperature.' Do NOT rush this drawing; give learners time to copy. For the Socratic Seminar: pose Question 1 and wait 15 seconds in complete silence before cold-calling. Accept all answers. After the first response, redirect: 'Does anyone want to add to that — or challenge it?' After 3 student contributions on Q1, pose Q2. For Q3 (the gut enzyme analogy), cold-call a learner who has been quiet today: 'Amara — you haven't shared yet — what do you think?' Affirm the attempt regardless of accuracy, then open to the group. For the written explanation: circulate and read over shoulders. If a learner's explanation omits the	Energy Diagram Annotation + Socratic Seminar — the energy profile diagram makes the abstract concept of activation energy concrete and visual, allowing learners to literally see the difference a catalyst makes. The Socratic Seminar then forces learners to transfer this visual model to two new contexts (light, enzymes), which is the hallmark of deep conceptual understanding. The sentence-frame writing task consolidates the transfer and creates an artefact for formative assessment.	Observable indicator: Learner's written 3-sentence explanation explicitly mentions 'activation energy' or 'energy pathway' AND connects to the thiosulfate phenomenon. Teacher collects 6 randomly selected notebooks (one per group) during the DQB phase to scan for this indicator. During the Socratic Seminar, monitor whether learners distinguish between a catalyst (lowers E_a, not consumed) and a reactant (consumed) — any conflation of these two is a misconception to address whole-class before Phase 4.

	started with in Lesson 1?' 3. 'Think about an enzyme in your gut breaking down ugali starch. Is that enzyme acting more like MnO_2 in our experiment today — or more like the warm water in Beaker B from our anchoring phenomenon? Explain your reasoning.' Learners write a 3-sentence explanation in their notebooks connecting today's results to the overarching driving question, using the sentence frame: 'In [context], [catalyst/light] increases the rate of reaction by [mechanism], which is similar to what happens in Beaker B of the thiosulfate experiment because…'	collision model, kneel to their level and whisper: 'Can you add ONE word about particles or energy to your explanation?' Do not rewrite for them.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Equivalent Rates (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes of different colours: – YELLOW sticky note: 'What we now KNOW about catalysts and light as rate factors (with evidence from today's experiment).' – BLUE sticky note: 'A NEW question today's experiment raised that we still cannot fully answer.' Groups write their notes (3 min) and one representative from each group walks to the class Driving Question Board and places both notes in the appropriate columns under the row labelled 'Lesson 7 — Catalysts & Light.' The class reads the board together (1 min silent reading), then the teacher facilitates a 4-minute whole-class discussion comparing the six groups' BLUE (new	Hand out sticky notes and set a visible 3-minute timer. Circulate to ensure YELLOW notes contain specific evidence ('our glowing splint relighted,' 'MnO_2 was still present after the reaction') rather than vague statements ('it was faster'). If a group's YELLOW note lacks evidence, crouch down and ask: 'What did you observe in your notebook that proves this? Add that detail.' At the board, after all notes are placed, stand back and say: 'Look at our DQB — we have now investigated FIVE factors: temperature, concentration, surface area, catalysts, and light. Read the blue question notes silently for 60 seconds.' (Wait the full 60 s.) Then ask: 'Which of these new questions do you think Lesson 8	Colour-Coded DQB Protocol — the two-colour sticky note system explicitly separates 'consolidated evidence' (YELLOW) from 'productive uncertainty' (BLUE), teaching learners that science generates new questions as fast as it answers old ones. The running Factor Summary Table functions as a cumulative sense-making artefact that makes the growing explanatory framework visible to learners across all eight lessons.	Observable indicator: Each group's YELLOW note contains at least one piece of specific experimental evidence (not just a claim). Teacher photographs the DQB after the lesson for ongoing documentation. The Factor Summary Table completion is checked during Phase 5 — any learner missing more than two rows from previous lessons receives a catch-up prompt card.

	question) notes, identifying common threads and unresolved puzzles. Learners update their personal 'Factor Summary Table' in their notebooks — a running table they have maintained since Lesson 3 — adding two new rows: 'Catalyst' and 'Light,' filling in columns for 'What changes,' 'Effect on collision frequency/energy,' 'Kenyan real-life example,' and 'Connection to thiosulfate phenomenon.'	will help us answer — and which might be questions that scientists in Kenya are STILL researching?' Cold-call 2 learners. Close with: 'Our driving question asks how we can control reaction speed to make life in Kenya safer, healthier and more productive. Look at your Factor Summary Table — you now have FIVE tools to answer that. On your way out, think about which tool is most important for a tea farmer in Kericho.'		
Model Building Phase 📹 **VIDEO:** [Common polyatomic ions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Equivalent Rates (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative particle-level models — the annotated diagrams they have been building since Lesson 2, showing what is happening INSIDE the beakers of the thiosulfate experiment at the particle level. Today they add two new annotations using a different pen colour (red for Lesson 7): 1. Catalyst annotation: They add a small labelled 'MnO_2-style catalyst particle' to their model and draw an arrow showing it providing an 'alternative lower-energy pathway' for collisions — WITHOUT disappearing from the diagram after the reaction. 2. Light annotation: They add a 'photon arrow' entering the beaker from above, with the label: 'photon → gives energy to particle → lowers the energy needed for a successful collision → increases rate.' Learners then write a one-paragraph 'Model Narrative' beneath their diagram: 'My model now shows that reaction rate can	Distribute the model portfolios (or ask learners to open their notebooks to the cumulative model page). Say: 'You are going to make TWO additions in red today — one for catalysts, one for light. Your model should now show ALL five factors we have investigated. Take 5 minutes.' Set timer. Circulate and look specifically for: (a) Is the catalyst particle still present AFTER the reaction in the diagram? If not, say quietly: 'Check your Station A data — was the MnO_2 still there at the end? Update your model.' (b) Is the photon arrow drawn as an energy INPUT — not as a new reactant? If confused, ask: 'Does light become part of the product — or does it just give energy to the reaction? How does that compare to what heat does in Beaker B?' With 2 minutes remaining, ask learners to write their final sentence ('The one question I most want Lesson 8 to answer…'). Collect 3–4 model portfolios to review overnight for misconceptions to address at the start of Lesson 8. Close by saying:	Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models) — requiring learners to physically annotate the SAME model page across all eight lessons makes the growth of their explanatory framework visible and concrete. The colour-coding by lesson allows both the learner and the teacher to see exactly what conceptual additions each lesson contributed. The Model Narrative paragraph forces learners to integrate all five factors into a single coherent explanation, which is the core cognitive demand of the unit.	Observable indicator: The catalyst annotation in the model shows the catalyst particle present both BEFORE and AFTER the reaction (i.e., not consumed); the light annotation shows a photon as an energy input. The Model Narrative mentions all five rate factors by name and includes the Kenyan real-world application. The final 'one question' sentence provides diagnostic data for Lesson 8 planning. Teacher reviews overnight and prepares a brief 3-minute 'misconception correction' slot for the start of Lesson 8 based on the 3–4 collected portfolios.

	be increased by [list all five factors], each of which works by [increasing collision frequency / increasing collision energy / lowering activation energy / providing energy directly to particles]. The thiosulfate cross disappears faster in Beaker B (warm, Mombasa conditions) because [link to model]. In a maize field in the Rift Valley, a farmer could [apply one factor] to [control a specific reaction rate process] because [mechanism].' At the very end, learners write one sentence at the bottom of their model page: 'The one question I most want Lesson 8 to answer is: _______________.'	'Lesson 8 is our synthesis lesson. You will use every factor on your Summary Table to write a full scientific explanation of the thiosulfate phenomenon AND design a reaction-rate control strategy for a real Kenyan context. Your models today are your revision notes. Guard them.'		

D. TEACHER REFLECTION

After the lesson, consider the following questions for your professional journal:\n1. CATALYST MISCONCEPTION CHECK: Did any learners persist in drawing the MnO_2 as 'used up' in their particle models? If so, was this corrected during Phase 5 or does it need to be addressed whole-class at the start of Lesson 8? Note the names of affected learners.\n2. LIGHT MECHANISM DEPTH: Did learners successfully distinguish between light as an energy source (parallel to heat) versus light as a reactant? The photosynthesis analogy in Kenyan maize and tea farming is a powerful bridge here — did you use it, and did learners connect it?\n3. SAFETY COMPLIANCE: Were all goggles and gloves worn for the full duration of Station A? If PPE compliance was poor, reflect on your briefing — consider assigning a permanent Safety Officer role in future lessons.\n4. STATION TIMING: Did both stations run for the full 20 minutes, or did one group finish early while another was still setting up? If timing was uneven, consider pre-setting the MnO_2 spatulas and pre-cutting the photosensitive strips before the lesson next time.\n5. DRIVING QUESTION PROGRESSION: Read through the BLUE sticky notes your learners posted on the DQB. Do the new questions they are raising show that they are thinking at a mechanistic, particle level — or are they still asking descriptive questions ('what happens if…')? The quality of their new questions is your best measure of the conceptual depth this lesson achieved.\n6. LESSON 8 READINESS: Based on the model portfolios you collected, identify the top two misconceptions to address in Lesson 8's opening. Write them here before you sleep.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In Station A, adding manganese dioxide (MnO_2) to hydrogen peroxide caused immediate vigorous effervescence and relighted a glowing splint, confirming oxygen production — the reaction was dramatically faster than the control with no MnO_2. Crucially, the MnO_2 powder was still visible on the filter paper after the reaction had ended, showing it was not consumed. In Station B, the photosensitive paper exposed to bright light or direct sunlight changed colour significantly faster than the strip kept in dim light, and the strip stored in the dark showed no change at all.
What did I learn?	A catalyst is a substance that increases the rate of a chemical reaction by providing an alternative reaction pathway with a lower

	activation energy, without being consumed in the process. This means that at the same temperature, more collisions have enough energy to result in a successful reaction. Light can also increase the rate of certain chemical reactions (photochemical reactions) by supplying energy directly to reactant particles — similar to the way heat energy increases collision energy in the thiosulfate experiment. Both catalysts and light therefore increase reaction rate through an energy mechanism, but neither changes the concentration or surface area of the reactants.
How does this explain the phenomenon?	All five rate factors investigated across Lessons 3–7 (temperature, concentration, surface area, catalysts, and light) increase reaction rate either by increasing the frequency of collisions between particles, by increasing the energy of those collisions, or by lowering the minimum energy (activation energy) that particles need for a collision to be successful. In the anchoring phenomenon, Beaker B (warm, ~50 °C, simulating Mombasa/Kisumu heat) reacted faster than Beaker A (cold, ~10 °C, simulating a cool Nairobi highland morning) because heat gave more sodium thiosulfate and HCl particles sufficient energy to collide successfully — the same principle as a catalyst or light providing the energy boost needed to overcome the activation energy barrier. In Kenya, enzymes in the human gut act as biological catalysts to break down ugali and githeri efficiently at body temperature; chlorophyll acts as a light-harvesting catalyst in photosynthesis in Rift Valley tea and maize crops; and industrial catalysts are used in fertiliser manufacturing to speed up the Haber process at Kenyan agro-chemical plants.

LESSON 8 (40 minutes): Model Building, Assessment, and Community Connection: Closing the Case on Reaction Rates	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students consolidate all four rate factors into a single unified particle-collision model, apply that model to novel Kenyan scenarios, close out the Driving Question Board, and connect reaction-rate science to real careers and community responsibilities in Kenya.
Knowledge	Students will be able to state and explain how temperature, concentration, surface area, and catalysts each affect the rate of a chemical reaction using collision theory; and articulate how these principles operate in at least two Kenyan biological or industrial contexts.
Skills	Students will construct, present, and evaluate a cumulative particle-collision model diagram; use evidence from all seven preceding lessons to justify claims; and apply their model to predict and explain outcomes in a novel written scenario.
Attitudes	Students will demonstrate scientific responsibility by honestly evaluating the strengths and limitations of their own models; show respect for peers during presentations; and appreciate how chemistry knowledge contributes to community wellbeing in Kenya.
Key Inquiry Question	How do various factors affect the rate of reactions — and how can controlling those factors solve real problems in Kenyan homes, industries, and health contexts?
Purpose in Storyline	This is the synthesis lesson that closes the eight-lesson arc. Students began by watching a cross disappear faster in a warm beaker and asking why. Every lesson since has added one piece of the explanation. Today they assemble all the pieces into one coherent model, demonstrate understanding through a novel-scenario assessment, and formally answer the driving question — connecting particle-level science to community-level impact across Kenya.
Safety Notes	No new hazardous chemicals are used in this lesson. If students handle any residual glassware from previous practicals, standard laboratory safety applies: wash hands after contact with any chemical residue, do not taste any substance, and report any breakage immediately. Emotional safety: ensure all group presentations are received respectfully; the teacher should model constructive feedback language before presentations begin.

B. LESSON OVERVIEW

Lesson 8 is the final consolidation lesson for Sub-strand 2.5. Students open by individually predicting how a novel Kenyan scenario (fish drying at Lake Victoria) would change if multiple rate factors were altered simultaneously — activating prior knowledge from all seven lessons. They then observe a brief recap gallery of the key data they generated across lessons (posted charts: thiosulfate cross times at four temperatures, concentration graphs, CO_2 volume curves, liver-and-peroxide results) and use that evidence to refine their predictions. In the Explain phase, groups present their finalised cumulative particle-collision models and receive structured peer feedback. The Driving Question Board is ceremonially closed: students vote on the most satisfying answer, identify remaining curiosities, and connect the science to Kenyan careers. A brief written predict-explain assessment using a novel scenario anchors individual accountability. The lesson ends with the teacher explicitly linking the disappearing-cross phenomenon observed on Day 1 to every factor and application studied — completing the storyline.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students receive a printed or	Distribute or display the scenario	Activation of prior knowledge	Teacher circulates during the 4-

📹 **VIDEO:** [Dividing to Find the Rate] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	board-displayed novel scenario card titled 'The Lake Victoria Fish Drier Problem': A fisherwoman at Lake Victoria in Kisumu wants to dry her dagaa (small fish) as quickly as possible to prevent spoilage. She has four options — (A) lay the fish whole on raised racks in open sunlight, (B) cut the fish into thin strips and lay on racks in open sunlight, (C) rub the fish with large-grain salt and store in a cool shed, or (D) place whole fish in a sealed container at room temperature. Students individually write in their notebooks: (1) Which option will slow spoilage the most? (2) Which will dry the fish fastest? (3) For each option, identify which rate factor is being changed and state whether it increases or decreases the reaction rate. Students are explicitly told to use the language of collision theory in at least one sentence. After 4 minutes of individual thinking, students share predictions with a shoulder partner for 1 minute before teacher takes responses.	card at the start of the lesson without preamble. Say: 'Before we look at anything from our past lessons, I want to see what YOU now know. Work alone for 4 minutes — no talking, no looking at notes — just use what is in your head. Write your prediction and explain it using collision theory.' Set a visible timer. After 4 minutes say: 'Turn to your shoulder partner. You each have 30 seconds to share your prediction. Your job is to find ONE point where you agree and ONE point where you disagree.' After the partner share, cold-call three pairs using equity sticks. Record the range of predictions on the board in two columns: Factor Identified and Direction of Effect. Do NOT evaluate correctness yet. Say: 'We will come back to these predictions after we look at the evidence we collected together across seven lessons. Keep your prediction visible in your notebook.' WAIT TIME: Allow a full 4-minute silent individual think before any partner talk. After cold-calling, pause 10 seconds after each student response before commenting or calling the next student.	through novel transfer task. The Lake Victoria dagaa scenario is deliberately unfamiliar as a context but directly applies all four factors studied. By requiring collision-theory language, the teacher assesses depth of conceptual transfer rather than recall of specific experimental results. Recording predictions publicly creates accountability and a reference point for the Explain phase.	minute silent write and scans notebooks for: correct identification of factor (surface area for option B, concentration/osmosis analogy for C, temperature for D, light and surface area for A); presence of collision-theory language; direction of effect (increase or decrease rate). Common gaps to note: students who identify the factor but cannot connect it to collision frequency; students who confuse preservation (slowing rate) with drying speed (different process). These gaps are addressed in the Explain phase presentations.
Observe Phase 📹 **VIDEO:** [Dividing to Find the Rate] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Formulas for Problem Solving: Finding Distance, Rate, and Time]	Students conduct a structured gallery walk around the classroom. Posted on the walls are six labelled evidence stations, each containing the key data chart or diagram produced during the preceding seven lessons: Station 1 — Thiosulfate cross disappearance times (cold vs. warm beaker, Lesson 1 anchoring observation); Station 2 — Rate vs. Temperature graph (Lesson 3 class data, showing increasing rate with temperature); Station 3	Set up the gallery stations before students arrive, or during the Predict phase if room permits. Before releasing students say: 'This is your own data — you collected it, you plotted it. Now use it as a scientist would: as evidence to test your ideas. At each station, add a sticky note with ONE sentence that starts: This evidence shows that... and includes the word collision. You have 5 minutes. Move efficiently.' Circulate during the gallery walk.	The gallery walk uses students own empirical data as the primary evidence source, reinforcing that scientific conclusions must be grounded in observation. Requiring collision-theory language on every sticky note compels students to make the mechanistic connection at each station rather than simply reading the graph. The star-marking activity directly links the observation phase back to their Lake Victoria predictions, creating	Teacher reads sticky notes added at each station during circulation and after the gallery walk. Look for: accurate connection of evidence to collision theory (frequency or energy of collisions); errors such as 'hotter temperature means stronger chemicals' rather than 'faster-moving particles collide more often'; students who connect multiple stations (e.g., noting that both temperature and catalysts work through an energy mechanism). Station 5 (liver-and-

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	— Rate vs. Concentration graph (Lesson 4 class data); Station 4 — CO2 volume-vs.-time curves for large chips, small chips, and powder (Lesson 5); Station 5 — Liver-and-peroxide results table (Lesson 6, fresh vs. boiled liver vs. sand); Station 6 — Annotated Kenyan context diagrams created by groups in Lesson 7 (rusting, fermentation, digestion, fish preservation). Each station has a sticky-note prompt: 'What does this evidence tell us about WHY reaction rate changes?' Students walk in pairs for 5 minutes, adding one sticky note of their own at each station with a single sentence connecting that evidence to collision theory. They also mark any station where the evidence changed or confirmed their Lake Victoria prediction with a star.	Pause at Station 1 (the original thiosulfate cross data) and ask quietly to passing pairs: 'This is where we started on Day 1. What do you understand now that you did not understand then?' — do not require a public answer yet, plant the reflection. At the 4-minute mark give a 1-minute warning. After the gallery walk, ask the whole class: 'Show me with fingers: how many stations confirmed your Lake Victoria prediction? How many challenged it?' Use the show of hands to gauge readiness for the Explain phase. WAIT TIME: At Station 1, after asking the quiet question to a pair, wait 8 seconds for a response before moving on.	a predict-observe-explain cycle within the broader lesson structure.	peroxide) is a known difficulty point — watch for students who can name the catalyst but cannot explain why boiling destroys it. Note these for targeted questioning in the Explain phase.
Explain Phase 📹 **VIDEO:** Dividing to Find the Rate Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each group (from Lesson 7 groups) has 90 seconds to present their finalised cumulative particle-collision model. The model must be a single diagram or annotated poster that shows: (a) a central particle-collision diagram with labelled successful and unsuccessful collisions; (b) four labelled arrows showing how temperature, concentration, surface area, and catalysts each change the collision frequency or energy; (c) at least two Kenyan contexts annotated on the diagram (e.g., cooking githeri in Kisumu, rusting of mabati roofs in Mombasa, KEMRI enzyme research in Nairobi, fermentation of uji or busaa); (d) a written answer to the driving question in one or two sentences. After each presentation, the class gives structured feedback using the	Before presentations begin, model the warm-wonder feedback protocol explicitly: say 'A warm sounds like: Your model clearly shows that... A wonder sounds like: I am curious about how your model explains... Practice those sentence starters — do not say good job or that is wrong.' Use a visible 90-second timer for each group presentation. After each warm-wonder round, the presenting group has 20 seconds to respond to the wonder — this is not a debate, just a brief clarification or acknowledgement. After all groups present, transition to the individual assessment by saying: 'You have seen all the evidence. You have heard all the models. Now I want to know what YOU understand — alone. This is your chance to show yourself and me how your thinking has grown	The group presentation externalises and makes the cumulative model a shared object of critique, not just a private notebook entry. The structured warm-wonder protocol teaches scientific discourse norms — that productive critique is respectful and question-driven. The individual written assessment creates personal accountability and provides the teacher with summative evidence of each students conceptual understanding. Placing the assessment after the gallery and presentations means students have activated all relevant knowledge before writing — this is not a test of memory under pressure but of reasoning under support.	Written assessment responses are collected and reviewed against the following success criteria: (1) correct identification of surface area as the key variable (Factory B powder dissolves faster); (2) collision-theory explanation — powder has greater surface area exposed, providing more sites where acid particles can collide with carbonate particles, increasing collision frequency and therefore rate; (3) a plausible and correctly justified additional change (e.g., heating the stomach acid solution — higher temperature increases kinetic energy of particles, more energetic collisions, more exceed activation energy; or increasing acid concentration — more acid particles per unit volume, more frequent collisions). Students who suggest adding a catalyst should

	protocol: one warm (what the model explains well) and one wonder (one question the model raises or one thing that is unclear). The teacher then administers the individual written assessment: a predict-explain task using a second novel scenario — 'A KEBS inspector finds that antacid tablets from two factories dissolve at very different speeds in the same stomach-acid solution at the same temperature. Factory A tablets are large and chalky. Factory B tablets are fine powder. Using collision theory, predict which dissolves faster and explain why at the particle level. Then suggest ONE additional change Factory A could make — other than crushing their tablet — to increase the dissolution rate, and explain your suggestion.' Students write individually for 6 minutes.	since Lesson 1 when you first saw that cross disappear.' Distribute the assessment prompt. Circulate silently during the 6-minute write — do not answer questions about the prompt, instead redirect: 'What did your model show you about that factor?' After 6 minutes collect written responses. WAIT TIME: After each group presentation, wait 8 full seconds of silence before inviting warm-wonder responses from the audience — resist the urge to fill the silence.		be asked in follow-up whether a safe edible catalyst exists — this extends thinking for advanced learners. Students who cannot connect their suggestion to collision theory have not yet achieved the SLO and require follow-up support.
Driving Question Board (DQB) Creation 📹 **VIDEO:** DNA as the "transforming principle" Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The Driving Question Board — maintained since Lesson 1 — is brought to the front of the class. Students silently review all the question cards posted over the eight lessons (approximately 2 minutes). The teacher then reads the original driving question aloud: 'Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive?' Each student receives two sticky dots: one green (for a question on the board they feel has been fully answered) and one orange (for a question they feel still deserves further investigation). Students place their dots on the board and the class briefly discusses the distribution. Three students are called on to	Retrieve the DQB prominently before students arrive or appoint a student group to set it up as the lesson begins. During the 2-minute silent review say: 'Read every card. Think about which ones you could now answer confidently using the language of collision theory.' After dot placement, use equity sticks to select the three green-dot reporters and two orange-dot reporters. For each green-dot response, follow up with: 'Can anyone add a Kenyan example that supports that answer?' For orange-dot questions, validate the uncertainty: 'That is a genuine scientific question — researchers at KEMRI or the University of Nairobi are likely asking something related right now. Write it in your notebook as a question for your future self.'	Closing the DQB is a metacognitive ritual that helps students recognise their own intellectual growth over eight lessons. The green-orange dot activity gives every student a voice in evaluating what the class has learned collectively, not just what the teacher says was learned. Requiring collision-theory language in the verbal answers ensures the closure is conceptual, not merely declarative. Acknowledging unanswered questions validates scientific humility and positions curiosity as a virtue rather than a failure.	Teacher observes dot placement patterns: if many questions receive no green dots, re-examine whether those concepts were sufficiently addressed in previous lessons and plan remediation. Listen to the three green-dot verbal responses for accuracy of collision-theory language — if students give surface-level answers ('temperature makes it faster'), prompt: 'Why does temperature make it faster — what is happening to the particles?' The orange-dot questions become a record of extensions and enrichment opportunities for the teacher to incorporate into future planning or to share with the school science club.

	read aloud the question they marked green and give a one-sentence answer using collision theory. Two students share the questions they marked orange and explain why the question remains open. The teacher formally closes the DQB by saying: 'Every question you asked on Day 1 came from one observation — a cross disappearing in a warm beaker. Today we close that board not because science is finished, but because you have built enough understanding to take the next step.' The board is photographed for the class portfolio.	End the DQB closure with a show of hands: 'Who feels they can now answer the driving question better than they could on Day 1?' Then: 'Who has MORE questions now than they did on Day 1?' Affirm both responses as signs of scientific growth. WAIT TIME: After asking 'Who can now answer the driving question better?' wait 5 seconds before asking for hands — allow genuine reflection, not automatic response.		
Model Building Phase 📹 **VIDEO:** [Dividing to Find the Rate] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	In the final 6 minutes, students individually open their science notebooks to a fresh page and write their personal final answer to the driving question in 3 to 5 sentences. They must include: the name of at least one Kenyan place or context; reference to particle collisions; mention of at least two rate factors; and one sentence connecting the science to a career or responsibility. Students then share their answer with a partner in a final 60-second paired read-aloud. The lesson closes with the teacher connecting the very first observation — the disappearing cross — to the driving question answer: 'On Day 1 you watched a cross disappear and did not know why. Today you know that the warm beaker made sulfur particles and thiosulfate particles collide more frequently and with more energy — enough energy to react — and that same principle explains why a KEMRI scientist controls the temperature of an enzyme reaction, why a fish trader in Kisumu dries dagaa in the sun,	Announce the personal writing task clearly: 'Last task — and this one is yours alone. Open a new page. Write your final answer to the driving question. You have 4 minutes. Include a Kenyan place, particle collisions, two rate factors, and one sentence about a career or responsibility. Go.' Set a timer. Circulate and read over shoulders — offer one specific encouragement per student where possible: 'I like how you used the word activation energy there' or 'Can you name the specific Kenyan place you are thinking of?' After 4 minutes, say: 'Read your answer to your partner. You each have 30 seconds. GO.' After the paired share, deliver the closing speech slowly and deliberately — this is the narrative payoff of the entire unit. Speak the names of real Kenyan institutions (KEMRI, Kenyatta National Hospital, Athi River, Brookside Dairy) with emphasis. End by saying: 'The cross has disappeared — but your understanding has not. It is permanent.' Pause for 5 seconds	The personal written answer is the culminating sense-making product of the entire unit — it requires students to synthesise across eight lessons, multiple phenomena, and multiple Kenyan contexts into their own words. The paired read-aloud gives an immediate low-stakes audience and often prompts spontaneous revision as students hear their own writing. The teacher closing narrative explicitly traces the arc from Day 1 observation to Day 8 explanation, modelling how scientists build understanding iteratively. Naming specific Kenyan institutions and career pathways makes the science personally relevant and vocationally motivating.	Collect or photograph personal written final answers as the exit artefact. Evaluate against four criteria: (1) Kenyan place or context named and correctly connected to a rate factor; (2) collision theory used mechanistically (not just named); (3) two or more rate factors mentioned with correct direction of effect; (4) career or responsibility connection is substantive (not just 'I want to be a chemist' but 'a food scientist at Brookside Dairy controls temperature during fermentation to keep reaction rates safe and consistent'). Students who meet all four criteria have demonstrated full achievement of the SLO. Students who meet two or fewer criteria are flagged for early intervention in the next sub-strand and should receive a targeted review task before the end-of-strand assessment.

	why a Mombasa roofer paints iron sheets to slow rusting, and why your own digestive system works best at 37 degrees Celsius. You are now scientists who can explain AND control the invisible world of colliding particles.' The teacher names three career pathways connected to rates of reactions: food scientist working with Kenya Breweries or Brookside Dairy, industrial chemist at Athi River or Titanium mining operations, pharmacist or biochemist at KEMRI or Kenyatta National Hospital. Students record these in their notebooks. A final round of applause for the class completing the unit is encouraged.	before leading the applause. WAIT TIME: After the closing speech, hold 5 full seconds of silence before the applause — allow the narrative to land.		

D. TEACHER REFLECTION
After Lesson 8, reflect on the following questions before closing your planning record: (1) Did the novel-scenario assessment (Factory A vs Factory B antacid tablets and the Lake Victoria dagaa problem) reveal any persistent misconceptions — for example, students conflating surface area with concentration, or students who could identify the correct factor but could not explain it using collision theory? If so, which lessons in this sub-strand would benefit from redesign? (2) Were the group model presentations genuinely cumulative — did they incorporate all four rate factors — or did groups default to the one factor their group investigated? If presentations were narrow, consider assigning cross-group jigsaw activities earlier in the storyline (e.g., after Lesson 5) to build cross-factor fluency before the final lesson. (3) Did the DQB closure feel authentic to students? If most questions received no green dots, consider whether the DQB was actively used across lessons 2 through 7 or whether it became a decorative display. The DQB is most powerful when teachers return to it at the start of each lesson and explicitly mark questions as they are answered. (4) Were Kenyan contexts genuinely integrated into student reasoning or were they surface-level labels? The goal is for a student to be able to say not just 'rusting is an example of reaction rate' but 'in Mombasa, the high humidity and salt concentration in sea air increase the rate of iron oxidation because more water and salt particles are present per unit volume, increasing collision frequency with the iron surface — painting the roof reduces the surface area of iron exposed.' If students cannot reach that level of specificity, the Lesson 7 context-application lesson may need more structured scaffolding. (5) Consider sharing the career connection (KEMRI, Athi River, Brookside Dairy, Kenyatta National Hospital) with the school career guidance counsellor so that interested students can be connected to further information or mentors in those fields.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed the disappearing thiosulfate cross for the last time — this time through the lens of all four rate factors. We also observed each groups cumulative particle-collision model and the full set of evidence charts from all seven lessons displayed as a gallery.
What did I learn?	We learned that temperature, concentration, surface area, and catalysts all affect the rate of a chemical reaction by changing the frequency or energy of particle collisions — and that this single principle explains phenomena as varied as cooking githeri in

	Kisumu, drying dagaa at Lake Victoria, rusting of mabati roofs in Mombasa, enzyme activity in the human gut, and industrial reactions at Athi River.
How does this explain the phenomenon?	We explained why the cross disappeared faster in the warm beaker on Day 1: higher temperature gave thiosulfate and hydrochloric acid particles more kinetic energy, causing more frequent and more energetic collisions, meaning more collisions exceeded the activation energy and produced sulfur precipitate faster. We also explained the Lake Victoria dagaa scenario and the Factory A vs Factory B antacid scenario using the same collision-theory model — showing that one set of ideas can explain many different Kenyan realities.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.